



NOW PLAYING

# Design and behaviour guide

The music player, the hub, the Windows companion, the Android app and the Discord bot: one Aqua system, its rules, every screen, and how to keep it true.

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## PROFILE

snow-leopard-itunes-9

## SOURCE FINGERPRINT

4e43856fb479381b

## COVERAGE

45 surfaces · 10 journeys · 25 rules

## INTERACTIVE VERSION

<docs/design/styleguide.html>

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# Now Playing styleguide

snow-leopard-itunes-9 · 118 colours · two skins · 45 surfaces · source 4e43856fb479381b

A reconstruction of Apple's 2009–2010 interface, built as one design system and shared by every part of Now Playing: the music player (a page, also inside the Android app), the hub's admin GUI and the Windows companion (both windows), and the words the Discord bot uses. Every value here is read from `tokens.json` as this page is built, every control on it is the real component wearing the real stylesheet, and every rule, screen and journey is read from `design/` — so the page cannot describe one thing and show another.

## The rules that outrank the visuals

A styleguide that opens with colour teaches the wrong lesson. These four decide whether a control is drawn at all, and each has overruled the aesthetics here at least once.

UX-PRIN-001

### A control that cannot act is not drawn

No button that looks live and does nothing, no toggle whose state is decoration. If an action is impossible in this context the control is omitted or replaced by an explained state. The equalizer window ships without traffic lights: three circles that close nothing would be decoration.

UX-PRIN-002

### An unavailable capability says so where the choice is made

Not in a tooltip, not in a log, not by disappearing. The control stays visible and, when chosen, reports the reason instead of acting. The Shared segment of the mode switch stays visible with no hub paired and reports why when pressed.

UX-PRIN-003

### Say what the code does, not what the feature is called

Labels describe the implemented behaviour. When a label and the DSP (or the network, or the provider) disagree, the label changes. Solfeggio presets are described as filters with no claimed physical effect; retuning states that the fallback changes tempo.

UX-PRIN-004

### A disabled control is still legible about why

Greyed, not gone. The reason is part of the accessible name, not only a tooltip. “500 Hz band, not used by 528 Hz (MI)”.

## Brand and voice

Now Playing is a self-hosted music suite: a player that works on the files you already have, a hub you run yourself, and a companion that brings a Windows library to it. Its look is a careful reconstruction of Apple's 2009–2010 interface — the Snow Leopard window, the iTunes 9 list, the iPod's Now Playing screen — chosen for its density and its honesty about state. The reference is `docs/design/APPLE_AQUA_2009_2010_UI_DESIGN_SPEC.md`.



### Now Playing Player

music-player/public/icon.svg

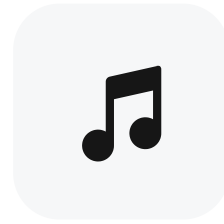
Also the Android launcher (brand colour #3b74c8) and the maskable PWA icon.



### Now Playing Companion for Windows

windows-companion/resources/icon.svg

The same tile with a folder behind the note: the one app that reaches the filesystem.



### The note

packages/aqua-ui/src/icons/glyphs.tsx

Both icons draw this exact path; a unit test fails if they drift apart again.

## USING THE MARK

- The tile is a rounded square (radius 112 on a 512 grid), a three-stop blue gradient and a soft gloss over the top half. Keep all three; the gloss is the period detail.
- Never recolour the note or redraw it by hand. Use the SVG or the `note` glyph.
- The blue tile is the product's mark. Inside the interface blue is never decoration — it means selected, active or default (see Colour).
- ● **PROPOSED** Clear space and minimum size have not been specified. Until they are, keep a quarter of the tile's width clear around it and do not use it below 16px; the tray icon is its own simplified drawing.

## VOICE

Plain, specific and a little dry. The product explains what it is doing and what it cannot do, in the place you are looking. It does not apologise, cheer or use exclamation marks, and it never claims more than the code does (UX-PRIN-003).

### VOICE-001

#### Empty states say what to do next ● OBSERVED

Title names the situation in a few plain words ("Nothing queued"); text gives the next action or the reason. No apologies, no exclamation marks. "Nothing queued — Play something from your library and it will appear here."

### VOICE-002

#### Errors explain the problem and the way out ● ADOPTED

Name what failed in the person's terms and what fixes it ("Add a bot token first (Admin → Discord → Token)"). Never show a stack trace or an internal identifier; a 500 carries a correlation id instead.

### VOICE-003

#### One product name, set in one place ● ADOPTED

"Now Playing" and the product names come from BRANDING in TypeScript. Native resources that cannot import it (Android strings.xml, the SVG icons' aria-labels, the Electron builder config) repeat it and must be changed together.

## WORDS WE USE

| SAY                            | MEANING                                                               | NOT                                     |
|--------------------------------|-----------------------------------------------------------------------|-----------------------------------------|
| <b>Now Playing</b>             | The suite. Its parts are “the player”, “the hub” and “the companion”. | Other product names invented per screen |
| <b>Solo / Shared listening</b> | The two modes on the status bar.                                      | Party mode, sync mode                   |
| <b>Group</b>                   | People listening to one shared queue on a hub.                        | Room, session                           |
| <b>Pair</b>                    | Connecting a device to a hub, confirmed by a fingerprint.             | Log in, register a device               |
| <b>Shared link</b>             | A public link to a track or playlist the hub serves.                  | Share URL, public page                  |
| <b>Fetch</b>                   | Running yt-dlp or spotDL on this machine.                             | Rip, grab                               |
| <b>Hub library</b>             | Files that live on the hub’s data volume.                             | Cloud, server music                     |

● OBSERVED The vocabulary above is how the code and the docs already talk; it has not been reviewed as a glossary.

## REAL COPY

### Nothing queued

Play something from your library and it will appear here.

Player · Queue

### No folders yet

Add the folder your music is in. The companion indexes it in place; your files are never copied or moved.

Companion · Folders

### Nothing to show yet

Play some music and this fills in. Everything here is worked out on this device from a log that never leaves it.

Player · Listening history

### Search your music

Searches the music on this device. Pair a hub in Settings to search connected services too.

Player · Search

## Two skins, one system

The same profile in two arrangements. A window frame drawn inside a browser viewport is a picture of a window rather than a window, and a 196 px source list is simply unavailable on a phone — so the player is a page while the desktop products stay windows. Controls are shared verbatim between them.

### Window skin

--aqua-\* · aqua.css, aqua-window.css, aqua-media.css · Lucida Grande · hub admin GUI, Windows companion

### Page skin

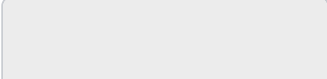
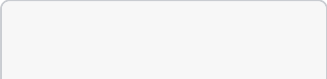
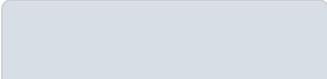
--np-\*, --lib-\* · now-playing.css · Helvetica, the iPod's face · music player PWA

# Colour

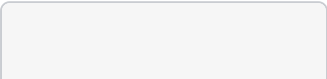
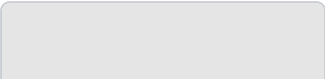
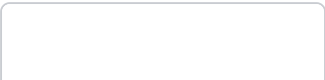
Chosen by role, never by picking a hex. Blue means *selected*, *active*, or *the default action* — it is not a brand colour and it is never decoration. Two palettes follow, straight from the token file: the window skin's, then the page skin's.

## WINDOW SKIN — 91 TOKENS

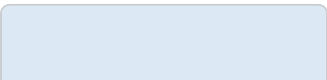
### WINDOW AND CONTENT

|                                                                                                              |                                                                                                               |                                                                                                             |                                                                                                           |
|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <br>desktop<br>#6D737A      | <br>windowOutline<br>#747474 | <br>windowBody<br>#ECECEC | <br>content<br>#FFFFFF |
| <br>contentMuted<br>#F7F7F7 | <br>rowStripe<br>#F1F6FB     | <br>rowDivider<br>#D8DEE5 |                                                                                                           |

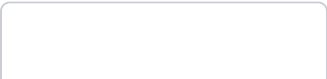
### CHROME

|                                                                                                                   |                                                                                                                   |                                                                                                               |                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <br>chromeTop<br>#F6F6F6        | <br>chromeUpper<br>#E5E5E5      | <br>chromeLower<br>#C4C4C4 | <br>chromeBottom<br>#AFAFAF |
| <br>chromeSeparator<br>#858585 | <br>chromeHighlight<br>#FFFFFF |                                                                                                               |                                                                                                                 |

### SOURCE LIST

|                                                                                                              |                                                                                                                 |                                                                                                                   |                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <br>sidebarTop<br>#DCE8F4 | <br>sidebarBottom<br>#BCEEE0 | <br>sidebarDivider<br>#93A3B2 | <br>sidebarGroupText<br>#536473 |
|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|

### SELECTION AND FOCUS

|                                                                                                                 |                                                                                                                |                                                                                                                    |                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <br>selectionTop<br>#69B5F3  | <br>selectionMid<br>#378DDA | <br>selectionBottom<br>#1764B2 | <br>selectionBorder<br>#0F4E8B |
| <br>selectionText<br>#FFFFFF | <br>focus<br>#3F9FE8        |                                                                                                                    |                                                                                                                     |

AQUA GEL

aquaSpecular  
#EAFBFF

aquaTop  
#A9E5FF

aquaMid  
#38B6F7

aquaLower  
#0D83D6

aquaBottom  
#075FA6

aquaRim  
#07558F

GRAPHITE

graphiteTop  
#E5E7E9

graphiteMid  
#AEB4B9

graphiteBottom  
#737A80

TEXT

text  
#161616

textSecondary  
#565B60

textDisabled  
#96999C

LCD

lcdTop  
#E1EAE8

lcdBottom  
#AEBCC0

lcdText  
#172326

STATUS

danger  
#D64A44

warning  
#D9A431

success  
#4E9D47

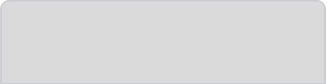
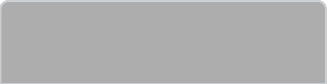
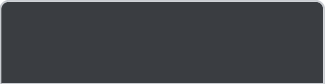
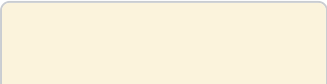
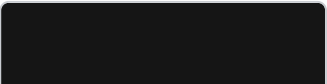
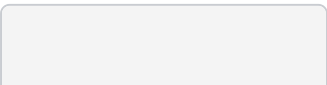
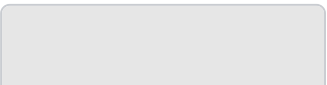
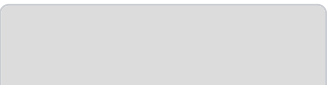
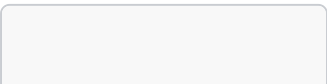
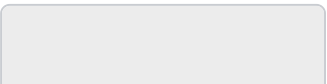
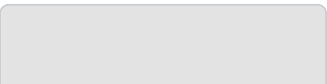
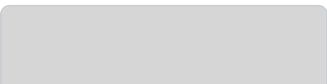
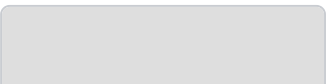
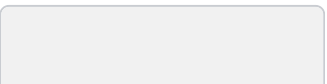
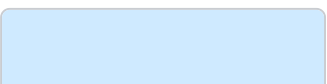
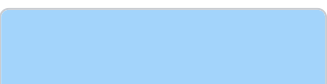
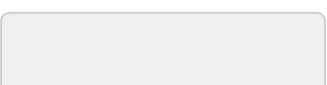
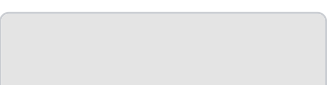
TRAFFIC LIGHTS

trafficClose  
#E9635B

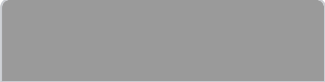
trafficMinimize  
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trafficZoom  
#58A953

## OTHER

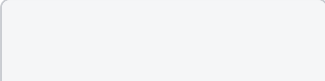
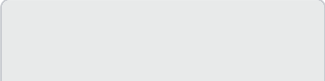
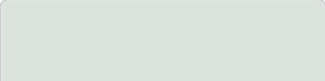
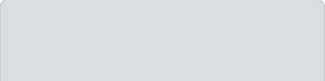
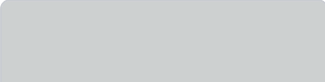
|                                                                                                                        |                                                                                                                           |                                                                                                                                  |                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <br><b>headerTop</b><br>#F8F8F8       | <br><b>headerBottom</b><br>#DADADA       | <br><b>headerRule</b><br>#ADADAD               | <br><b>headerText</b><br>#3A3D41       |
| <br><b>warningBg</b><br>#FBF3DC       | <br><b>warningEdge</b><br>#D8B75E        | <br><b>ctlInk</b><br>#151515                   | <br><b>ctlFace1</b><br>#FDFDFD         |
| <br><b>ctlFace2</b><br>#F4F4F4        | <br><b>ctlFace3</b><br>#E6E6E6           | <br><b>ctlFace4</b><br>#DCDCDC                 | <br><b>ctlFaceHover1</b><br>#FFF       |
| <br><b>ctlFaceHover2</b><br>#F8F8F8   | <br><b>ctlFaceHover3</b><br>#ECECEC      | <br><b>ctlFaceHover4</b><br>#E3E3E3            | <br><b>ctlFaceActive1</b><br>#C9C9C9   |
| <br><b>ctlFaceActive2</b><br>#D6D6D6  | <br><b>ctlFaceActive3</b><br>#DEDEDE     | <br><b>ctlFaceDisabled1</b><br>#FBFBFB         | <br><b>ctlFaceDisabled2</b><br>#F1F1F1 |
| <br><b>ctlBorder</b><br>#A3A3A3     | <br><b>ctlBorderBottom</b><br>#8D8D8D  | <br><b>ctlBorderDisabled</b><br>#C9C9C9      | <br><b>ctlFocusRing</b><br>#0D61A7   |
| <br><b>ctlDefaultInk</b><br>#06203A | <br><b>ctlDefault1</b><br>#CFEAFB      | <br><b>ctlDefault2</b><br>#A3D4FB            | <br><b>ctlDefault3</b><br>#74B9F5    |
| <br><b>ctlDefault4</b><br>#5AA8EF   | <br><b>ctlDefaultBorder</b><br>#4684C8 | <br><b>ctlDefaultBorderBottom</b><br>#356DB0 | <br><b>ctlBoxInk</b><br>#1A2733      |
| <br><b>ctlBox1</b><br>#FDFDFD       | <br><b>ctlBox2</b><br>#F0F0F0          | <br><b>ctlBox3</b><br>#E4E4E4                | <br><b>ctlBoxBorder</b><br>#A0A0A0   |
| <br><b>ctlBoxOn1</b><br>#7DB4E6     | <br><b>ctlBoxOn2</b><br>#4F92D6        | <br><b>ctlBoxOn3</b><br>#3F81C6              | <br><b>ctlBoxOnBorder</b><br>#3D76AD |



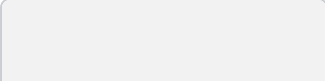
|                                                                                   |                                                                                   |                                                                                    |                                                                                     |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |
| ctlFieldBg<br>#FFF                                                                | ctlFieldBorder<br>#9A9A9A                                                         | ctlFieldBorderTop<br>#7D7D7D                                                       | ctlFieldFocusBorder<br>#1768A7                                                      |
|  |  |  |                                                                                     |
| ctlFieldInvalidBorder<br>#B8433D                                                  | ctlPlaceholder<br>#8B9095                                                         | dangerInk<br>#7A1712                                                               |                                                                                     |

## PAGE SKIN — 25 TOKENS

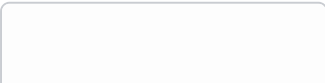
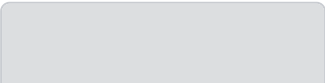
### STATUS BAR

|                                                                                   |                                                                                   |                                                                                    |                                                                                     |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |
| barTop<br>#F5F6F7                                                                 | barUpper<br>#E8EAEA                                                               | barLower<br>#DCE2DE                                                                | barBottom<br>#D9DFE1                                                                |
|  |                                                                                   |                                                                                    |                                                                                     |
| barEdge<br>RGB(31 41 45 / 22%)                                                    |                                                                                   |                                                                                    |                                                                                     |

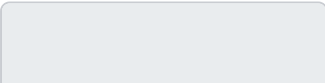
### FIELD

|                                                                                     |                                                                                     |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |
| fieldTop<br>#F2F2F2                                                                 | fieldBottom<br>#FFFFFF                                                              |

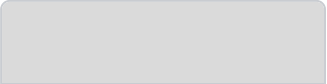
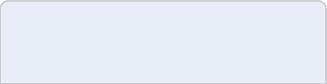
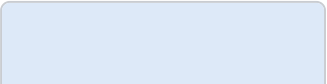
### MODE SWITCH

|                                                                                     |                                                                                     |                                                                                      |                                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  |  |  |  |
| modeFaceTop<br>#FDFDFD                                                              | modeFaceBottom<br>#DCDEE0                                                           | modeOnTop<br>#767C85                                                                 | modeOnBottom<br>#A9B0B8                                                               |

### SCRUBBER

|                                                                                     |                                                                                     |                                                                                      |                                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  |  |  |  |
| railTop<br>#E9ECEE                                                                  | railBottom<br>#FFFFFF                                                               | railFillTop<br>#8FB9E4                                                               | railFillMid<br>#2B9BEC                                                                |
|  |                                                                                     |                                                                                      |                                                                                       |
| railFillBottom<br>#A5E9FD                                                           |                                                                                     |                                                                                      |                                                                                       |

## LIST

|                                                                                                               |                                                                                                                  |                                                                                                                |                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <br>listHeaderTop<br>#F8F8F8 | <br>listHeaderBottom<br>#DADADA | <br>listSortTop<br>#E8EEF7   | <br>listSortBottom<br>#CFDBEB   |
| <br>listStripe<br>#F2F5F9    | <br>listPlaying<br>#DDE9F8      | <br>listSelectTop<br>#5B8DD9 | <br>listSelectBottom<br>#3A6CC4 |

## MARKS

|                                                                                                      |                                                                                                      |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <br>live<br>#CC1027 | <br>star<br>#F0A422 |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|

## DARK

The page skin has a full dark palette and it is opt-out: every dark rule is guarded `@media (prefers-color-scheme: dark) { :root:not([data-np-theme='light']) { ... } }`, so the system preference wins unless a page pins itself light. Redefine *only* tokens inside that block. The window skin has no dark palette, because Snow Leopard had none.

The page skin's colours that change in the dark scheme, read from the shipped stylesheet when this page was built (88 properties). Light on the left, dark on the right.

|                                                                |                                                                                                                                                                         |                                         |                                                                                                                                                                             |
|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --ctx-bg<br>rgb(247 247 250 / 88%) → rgb(40 42 48 / 90%)       |   | --lib-head-hot-bot<br>#cfcfcf → #303338 |   |
| --ctx-ink<br>#1b1c1f → #eef0f3                                 |   | --lib-head-hot-top<br>#efefef → #45484f |   |
| --ctx-sep<br>rgb(0 0 0 / 12%) → rgb(255 255 255 / 12%)         |   | --lib-head-ink<br>#3a3d41 → #c6cbd3     |   |
| --lib-accent<br>#2b6fd6 → #6aa3f2                              |   | --lib-head-rule<br>#adadad → #0f1114    |   |
| --lib-alt<br>#f2f5f9 → #1b1f26                                 |   | --lib-head-sep<br>#ccc → #17191d        |   |
| --lib-bg<br>#fff → #16181c                                     |   | --lib-head-top<br>#f8f8f8 → #3a3d44     |   |
| --lib-border<br>#ababab → #08090b                              |   | --lib-icon<br>#8a8f98 → #8d939c         |   |
| --lib-head-bot<br>#dadada → #26292e                            |   | --lib-ink<br>#000 → #f2f3f5             |   |
| --lib-head-emboss<br>rgb(255 255 255 / 80%) → rgb(0 0 0 / 55%) |   | --lib-np<br>#dde9f8 → #22354f           |   |

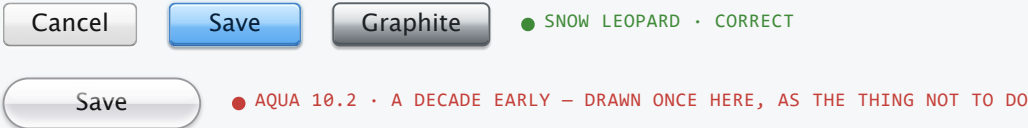
|                                                                    |                                                                                     |                                                                  |                                                                                       |
|--------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| --lib-soft<br>#222326 → #d4d8df                                    |    | --np-bar-top<br>#f5f6f7 → #34373d                                |    |
| --lib-sort-bot<br>#cfdbeb → #233043                                |    | --np-field-bottom<br>#fff → #24272d                              |    |
| --lib-sort-hot-bot<br>#c2d2e6 → #2b3b52                            |    | --np-field-hint<br>#ababab → #7c8087                             |    |
| --lib-sort-hot-top<br>#dde7f4 → #36485f                            |    | --np-field-icon<br>#8e8e91 → #85898f                             |    |
| --lib-sort-top<br>#e8eef7 → #2c3a4e                                |    | --np-field-lip<br>rgb(255 255 255 / 70%) → rgb(255 255 255 / 8%) |    |
| --lib-star<br>#f0a422 → #f4b241                                    |    | --np-field-mid<br>#fbfbfb → #1b1e23                              |    |
| --np-accent<br>#2b6fd6 → #6aa3f2                                   |    | --np-field-ring<br>rgb(0 0 0 / 20%) → rgb(0 0 0 / 70%)           |    |
| --np-avatar-face<br>#f3f3f3 → #2b2e34                              |    | --np-field-shadow<br>rgb(0 0 0 / 32%) → rgb(0 0 0 / 75%)         |    |
| --np-avatar-lip<br>rgb(255 255 255 / 70%) → rgb(255 255 255 / 10%) |   | --np-field-top<br>#f2f2f2 → #141619                              |   |
| --np-avatar-ring<br>rgb(0 0 0 / 26%) → rgb(0 0 0 / 55%)            |  | --np-ink<br>#1b1c1f → #f1f2f4                                    |  |
| --np-bar-1<br>#eef0f1 → #2f3238                                    |  | --np-key-ink<br>#1b1c1f → #ccd1d8                                |  |
| --np-bar-2<br>#e8eaea → #2a2d33                                    |  | --np-key-muted<br>#8b9099 → #7f858e                              |  |
| --np-bar-3<br>#e2e6e5 → #25282e                                    |  | --np-key-on<br>#2f6ec0 → #5fa0ea                                 |  |
| --np-bar-4<br>#dfe4e1 → #212429                                    |  | --np-list-alt<br>#f2f5f9 → #1b1f26                               |  |
| --np-bar-5<br>#dce2de → #1e2126                                    |  | --np-list-badge<br>#dfe3e9 → #2f333b                             |  |
| --np-bar-bottom<br>#d9dfe1 → #1a1d22                               |  | --np-list-badge-ink<br>#3b3f47 → #d6dae1                         |  |
| --np-bar-edge<br>rgb(31 41 45 / 22%) → rgb(0 0 0 / 60%)            |  | --np-list-border<br>#ababab → #08090b                            |  |
| --np-bar-emboss<br>rgb(255 255 255 / 55%) → transparent            |  | --np-list-icon<br>#8a8f98 → #8d939c                              |  |

|                                                |                                                                                                                                                                         |                                                |                                                                                                                                                                             |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| --np-live                                      |       | --np-vol-face                                  |       |
| #cc1027 → #ff4d62                              |                                                                                                                                                                         | #fff → #1b1e23                                 |                                                                                                                                                                             |
| --np-live-halo                                 |       | --srch-art-bot                                 |       |
| rgb(204 16 39 / 32%) → rgb(255 77 98 / 36%)    |                                                                                                                                                                         | #e3e3e3 → #24262b                              |                                                                                                                                                                             |
| --np-meta                                      |       | --srch-art-ink                                 |       |
| #575757 → #9ba0a8                              |                                                                                                                                                                         | #a9a9a9 → #8d939c                              |                                                                                                                                                                             |
| --np-page                                      |       | --srch-art-ring                                |       |
| #fff → #0e0f12                                 |                                                                                                                                                                         | rgb(0 0 0 / 14%) → rgb(255 255 255 / 10%)      |                                                                                                                                                                             |
| --np-rail-line                                 |       | --srch-art-top                                 |       |
| rgb(0 0 0 / 16%) → rgb(0 0 0 / 72%)            |                                                                                                                                                                         | #fbfbfb → #33363d                              |                                                                                                                                                                             |
| --np-seg-border                                |       | --srch-bg                                      |       |
| rgb(0 0 0 / 36%) → rgb(0 0 0 / 70%)            |                                                                                                                                                                         | #f9f9f9 → #26282e                              |                                                                                                                                                                             |
| --np-seg-divider                               |       | --srch-border                                  |       |
| rgb(0 0 0 / 30%) → rgb(0 0 0 / 55%)            |                                                                                                                                                                         | rgb(0 0 0 / 25%) → rgb(0 0 0 / 65%)            |                                                                                                                                                                             |
| --np-seg-face-bottom                           |       | --srch-chev                                    |       |
| #dcdee0 → #292c31                              |                                                                                                                                                                         | #8f8f8f → #9aa0a9                              |                                                                                                                                                                             |
| --np-seg-face-mid                              |     | --srch-dot                                     |     |
| #f0f1f2 → #33363c                              |                                                                                                                                                                         | #c9c9c9 → #4a4e56                              |                                                                                                                                                                             |
| --np-seg-face-top                              |   | --srch-dot-on                                  |   |
| #fdfdfd → #3a3d44                              |                                                                                                                                                                         | #7c7c7c → #b8bdc5                              |                                                                                                                                                                             |
| --np-seg-ink                                   |   | --srch-head-bot                                |   |
| #43464b → #c3c7cf                              |                                                                                                                                                                         | #eee → #25272d                                 |                                                                                                                                                                             |
| --np-seg-lip                                   |   | --srch-head-top                                |   |
| rgb(255 255 255 / 65%) → rgb(255 255 255 / 8%) |                                                                                                                                                                         | #f6f6f6 → #2e3138                              |                                                                                                                                                                             |
| --np-seg-on-bottom                             |   | --srch-hot                                     |   |
| #a9b0b8 → #23262c                              |                                                                                                                                                                         | #ececec → #32353c                              |                                                                                                                                                                             |
| --np-seg-on-mid                                |   | --srch-ink                                     |   |
| #939aa3 → #1c1f24                              |                                                                                                                                                                         | #2c2c2c → #eef0f3                              |                                                                                                                                                                             |
| --np-seg-on-top                                |   | --srch-lip                                     |   |
| #767c85 → #14161a                              |                                                                                                                                                                         | rgb(255 255 255 / 60%) → rgb(255 255 255 / 7%) |                                                                                                                                                                             |
| --np-vol-a                                     |   | --srch-rule                                    |   |
| #9aa1a9 → #6e747c                              |                                                                                                                                                                         | #dedede → rgb(0 0 0 / 50%)                     |                                                                                                                                                                             |
| --np-vol-b                                     |   | --srch-soft                                    |   |
| #b7bdc4 → #878e96                              |                                                                                                                                                                         | #9b9b9b → #8d939c                              |                                                                                                                                                                             |

## Aqua highlights — where the blue is allowed to shine

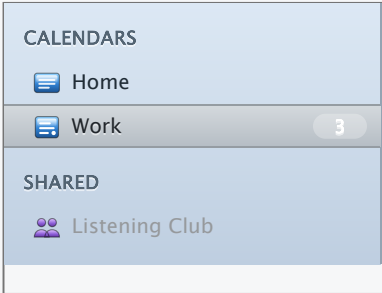
Snow Leopard had flattened most of Aqua's gel by 2009, but not all of it: selection, the default action, the scroller and a few controls kept the glass. Those are exactly the places this system keeps it, and this is each one, live.

### THE DEFAULT ACTION



The one button on a form that carries the Aqua blue: a pale cyan cap over a saturated body, a specular hairline at the top, a dark rim, and a slow pulse. The shape stays the Snow Leopard rounded rect — no pill, no glass lozenge, no gloss step across the middle.

### SELECTION



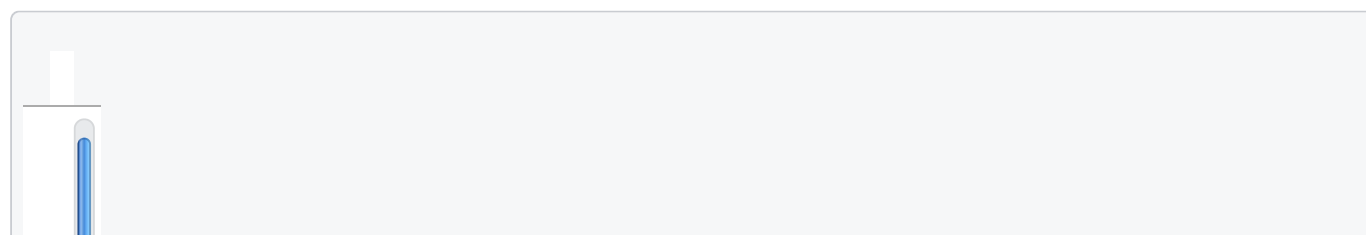
The source list's selected row is the blue gel gradient with white text, as in every OS X sidebar; the music list's selected row is the same blue at the page skin's tint; the sorted column is the palest tint of the three.

### CONTROLS THAT KEEP THE GEL



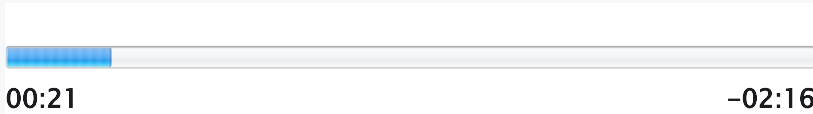
The checked box, the pop-up's end cap, the aqua-tinted segmented control and the indeterminate progress bar's barber pole are the last four places a form is blue.

### THE OVERLAY SCROLLER



The full five-stop Aqua gel survives at strength in exactly one place: the list's overlay scroller, which fades in while you scroll and out 900 ms after you stop.

## THE IPOD RAIL



Not gel, but the other blue on the page: the iPod Classic's progress bar, 12 px, near-square ends, and a fill that brightens downward — sky at the top through a saturated dip at 62% to a cyan glow along the lower lip. No knob: the device had none.

And the focus ring — `--aqua-focus #3F9FE8` with a soft outer glow — is blue everywhere, because the thing that has the keyboard is the one thing on screen that must never be ambiguous.

## Type

Two families, one scale. The window skin sets Lucida Grande, as OS X did. The page skin sets **Helvetica** — the face the iPod classic drew its Now Playing screen in, which is what the hero is modelled on — with Arial as its metric-compatible stand-in where Helvetica is absent. Each specimen is at its true size.

### PAGE SKIN · HELVETICA

|      |                                                 |                                 |
|------|-------------------------------------------------|---------------------------------|
| 13px | <b>Sundress — A\$AP Rocky</b>                   | <code>--aqua-font-system</code> |
| 12px | Marlow & the Tidewater — Quiet Arithmetic       | <code>--aqua-font-view</code>   |
| 11px | Searches this device only until a hub is paired | <code>--aqua-font-small</code>  |
| 10px | PREAMP · 32 · 64 · 125 · 250 · 500 · 1K · 2K    | <code>--aqua-font-label</code>  |
| 9px  | Mini controls only                              | <code>--aqua-font-mini</code>   |

### WINDOW SKIN · LUCIDA GRANDE

|      |                                         |                                 |
|------|-----------------------------------------|---------------------------------|
| 13px | <b>Now Playing</b>                      | <code>--aqua-font-system</code> |
| 12px | Copper Meridian — Orbital Cartographers | <code>--aqua-font-view</code>   |
| 11px | 1,240 songs, 2.6 hours, 245 MB          | <code>--aqua-font-small</code>  |
|      |                                         |                                 |

|      |                                 |                   |
|------|---------------------------------|-------------------|
| 10px | LIBRARY · PLAYLISTS · CONNECTED | --aqua-font-label |
| 9px  | Mini controls only              | --aqua-font-mini  |

Weights are **400** and **700**. There is no 500 and no 600: the profile predates variable UI fonts, and a half-weight reads as a rendering fault beside a true bold. Numbers that change in place — timers, dB values, tempo, track times — take font-variant-numeric: tabular-nums, so a changing digit does not shift the ones beside it.

## Space

One scale, no arbitrary values. Each bar is drawn at its true width.

|                                 |      |   |
|---------------------------------|------|---|
| --aqua-space-micro              | 2px  |   |
| --aqua-space-tight              | 4px  |   |
| --aqua-space-inline             | 6px  | ■ |
| --aqua-space-control            | 8px  | ■ |
| --aqua-space-small-group        | 10px | ■ |
| --aqua-space-group              | 12px | ■ |
| --aqua-space-frame-top          | 14px | ■ |
| --aqua-space-strong-group       | 16px | ■ |
| --aqua-space-window-edge        | 20px | ■ |
| --aqua-space-major              | 24px | ■ |
| --aqua-space-bottom-bar-regular | 32px | ■ |

### SIZES

|                              |                            |                       |                  |
|------------------------------|----------------------------|-----------------------|------------------|
| visualControlRegular<br>22PX | visualControlSmall<br>19PX | searchRegular<br>22PX | tableRow<br>20PX |
| listRow<br>18PX              | railHeight<br>12PX         | transportKey<br>48PX  | playKey<br>64PX  |
| barHeight<br>40PX            | sidebarDefault<br>196PX    |                       |                  |

**32 px** (--aqua-hit) is the floor for a pointer target, and **44 px** on a coarse pointer, which is what both platform guidelines ask for. A 12 px rail and an 18 px row are period-correct and stay that size — they get their hit area from an invisible ::after that expands past the visual. Never shrink a hit target to match a visual.

## Shape and material

One light source, above and slightly forward. Raised things are light at the top and dark at the bottom; recessed things are the reverse. Every shadow follows from that, and a rule that contradicts it looks broken even when the colours are right.

### RADII



## THE PUSH BUTTON

Add a folder...

Create group

Remove

Disabled

Small

Mini

A rounded rect, one quiet vertical gradient, a hairline rim that darkens along the bottom, one pixel of white on top. The default action is the same shape with the Aqua blue on it — see the highlights above.

```
background: linear-gradient(to bottom, #fdfdfd 0%, #f4f4f4 47%, #e6e6e6 53%, #dcdcdc 100%);
border: 1px solid #a3a3a3;
border-bottom-color: #8d8d8d;
border-radius: 4px;
box-shadow:
  inset 0 1px 0 rgb(255 255 255 / 90%), /* the pixel of light on the top edge */
  0 1px 0 rgb(255 255 255 / 55%);      /* the surface beneath catching it */
```

## RECESSED FIELDS

Q Search

Q Search your music

Wells run the other way: shadowed at the top inside, with the lower lip catching light. Same source, opposite surface. The window skin's search field and the page skin's search pill are the same idea at two heights (22 px and 26 px).

## FADERS

72 4 dB

The slider's knob is a small lozenge with a pointed edge, the equalizer's fader is the same control rotated; both ride a recessed groove.

## Motion

Confirmation, never decoration: nothing animates that a person did not just cause. Everything eases on `cubic-bezier(0.2, 0.7, 0.2, 1)`. Bars are drawn in proportion.

|                          |       |             |
|--------------------------|-------|-------------|
| --aqua-motion-press      | 70ms  | <div></div> |
| --aqua-motion-selection  | 100ms | <div></div> |
| --aqua-motion-disclosure | 140ms | <div></div> |



|                             |        |             |
|-----------------------------|--------|-------------|
| --aqua-motion-panel         | 200ms  | <div></div> |
| --aqua-motion-default-pulse | 1650ms | <div></div> |
| --aqua-motion-scroller-fade | 300ms  | <div></div> |

**Reduced motion is a contract, not a courtesy.** Under prefers-reduced-motion: reduce the live dot stops pulsing, the marquee does not scroll (the cell keeps its ellipsis), the sheet's travel drops to zero and it fades only, the constellation opens as a table rather than an animated field, and the 3D stage stops its idle drift. Two variables carry most of it — --aqua-anim-state: paused and --aqua-sheet-travel: 0 — so a new animation should read one of them rather than inventing a third.

## The page — what the player is made of

The music player's skin: a sticky status bar, a section strip, the hero with the jewel case on its stage, the iPod rail and a transport of eleven keys, and the iTunes 10 list under it. Everything below is live — press play and the case opens; pick a row and the stage follows it.

Now Playing

19:58

Search your music

Music Library

8

Queue

6

Playlists

Listening history

Midnight Set, Side B

Fennel Grove

Long Wave Sessions, Vol. 2

00:23

-02:35

| # |  | Song                 | Artist                 | Time | BPM |  |  | Album                      |
|---|--|----------------------|------------------------|------|-----|--|--|----------------------------|
| 1 |  | Copper Meridian      | Orbital Cartographers  | 5:05 | —   |  |  | Copper Meridian            |
| 2 |  | Ein außerordentl...  | Cassette Bloom         | 6:21 | 104 |  |  | Live from Pier 9           |
| 3 |  | Harbour Lights       | Orbital Cartographers  | 4:24 | —   |  |  | Copper Meridian            |
| 4 |  | Lantern Road         | Marlow & the Tidewater | 3:24 | 118 |  |  | Quiet Arithmetic           |
| 5 |  | Long Wave            | Fennel Grove           | 5:16 | 122 |  |  | Long Wave Sessions, Vol. 2 |
|   |  | Midnight Set, Side I | Fennel Grove           | 2:58 | 110 |  |  | Long Wave Sessions, Vol. 2 |
| 7 |  | Pier 9 (Live)        | Cassette Bloom         | 4:11 | —   |  |  | Live from Pier 9           |
| 8 |  | Quiet Arithmetic     | Marlow & the Tidewater | 3:51 | 96  |  |  | Quiet Arithmetic           |

**The transport is centred on the rail.** Play sits on the progress rail's centre line at every width, with previous and next either side of it on the same line, and the keys never run past the rail's ends. The volume line sits beside the keys only when it fits on both sides; otherwise it goes beneath, centred, as it does here. When the keys do not fit on one line, previous, play and next take the first line on their own and the other eight keys share the line below. The row measures itself to decide this, because the room it needs depends on how many keys a product carries (DEC-016).

THE MUSIC LIST

| # |  | Song                | Artist                 | Time | BPM |  | ★ | Album                      |
|---|--|---------------------|------------------------|------|-----|--|---|----------------------------|
| 1 |  | Copper Meridian     | Orbital Cartographers  | 5:05 | —   |  | ★ | Copper Meridian            |
| 2 |  | Ein außerorden...   | Cassette Bloom         | 6:21 | 104 |  | ★ | Live from Pier 9           |
| 3 |  | Harbour Lights      | Orbital Cartographers  | 4:24 | —   |  | ★ | Copper Meridian            |
| 4 |  | Lantern Road        | Marlow & the Tidewater | 3:24 | 118 |  | ★ | Quiet Arithmetic           |
| 5 |  | Long Wave           | Fennel Grove           | 5:16 | 122 |  | ★ | Long Wave Sessions, Vol. 2 |
| 6 |  | Midnight Set, Si... | Fennel Grove           | 2:58 | 110 |  | ★ | Long Wave Sessions, Vol. 2 |
| 7 |  | Pier 9 (Live)       | Cassette Bloom         | 4:11 | —   |  | ★ | Live from Pier 9           |
|   |  | Quiet Arithmetic    | Marlow & the Tidewater | 3:51 | 96  |  | ★ | Quiet Arithmetic           |

The same component the player renders, with the reference's stylesheet block: nine columns, 18 px rows, the stripe, no rules, an embossed header with the sorted column tinted, monochrome source badges read from each track's locator, icon columns for offline and star, the playing row in bold with a speaker glyph and a marquee, a right-click menu, and the gel scroller that fades in while you scroll.

THE SEARCH POPOVER

3 matches for "la"Clear

Lantern Road  
Marlow & the Tidewater — Quiet Arithmetic

3:24  
118 bpm

L

Long Wave  
Fennel Grove — Long Wave Sessions, Vol. 2

5:16  
122 bpm

YT

Copper Meridian  
Orbital Cartographers

5:05  
—

H

<

•

•

•

>

The iTunes 11 results popover: a gradient header with the count and Clear, 40 px rows whose artwork tile is a fifteen-second audition button with a countdown ring, a circled chevron on the hot row, and a pager. Arrow keys walk the rows and page off the end.

CONTEXT MENU, SHEET AND TOAST

Add to Playlist>

New Playlist...

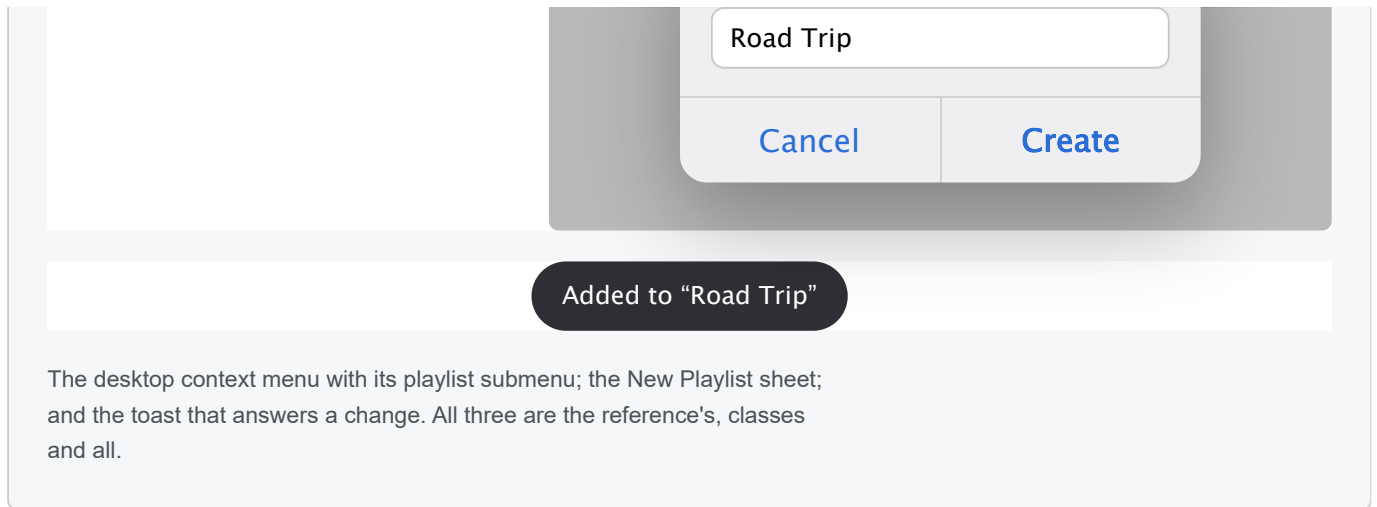
✓ Road Trip

Late Nights

New Playlist...

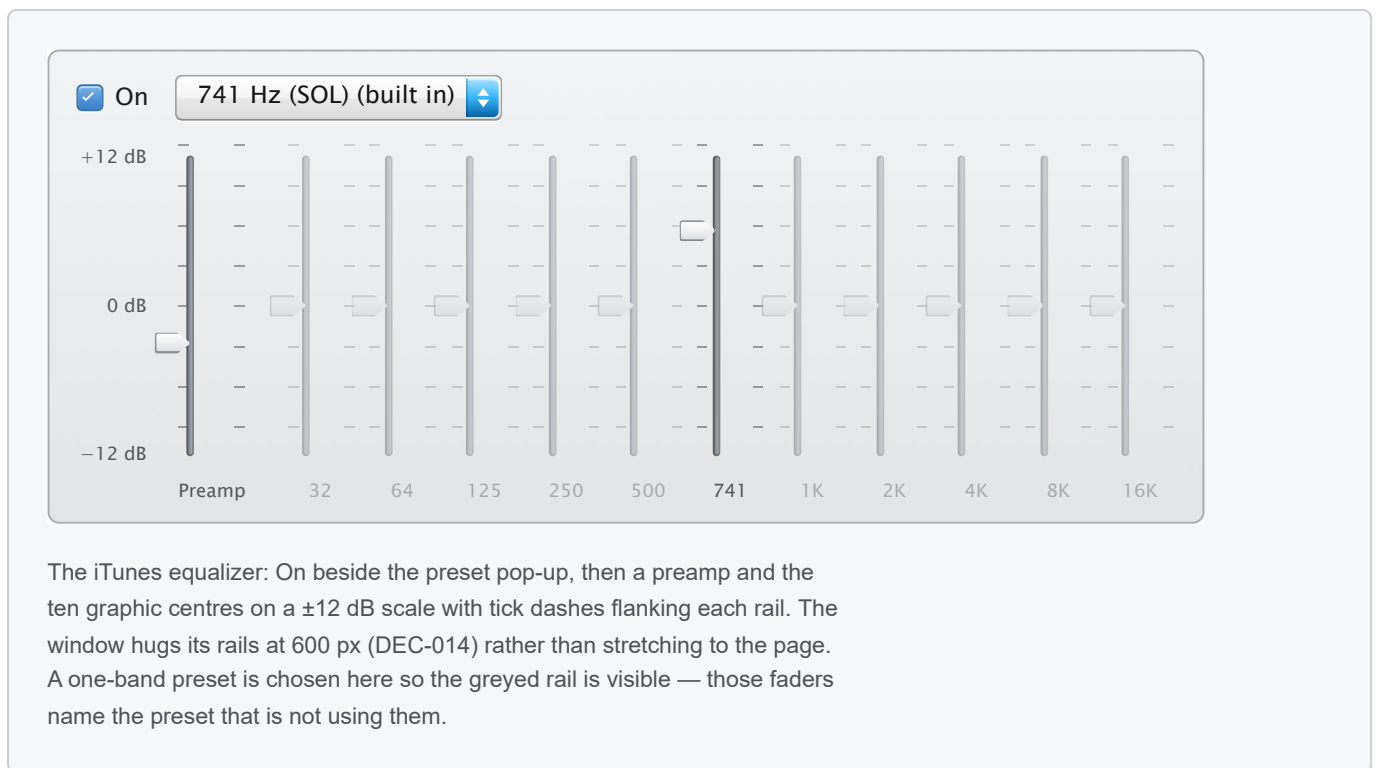
New Playlist

Enter a name. "Lantern Road" goes in it.



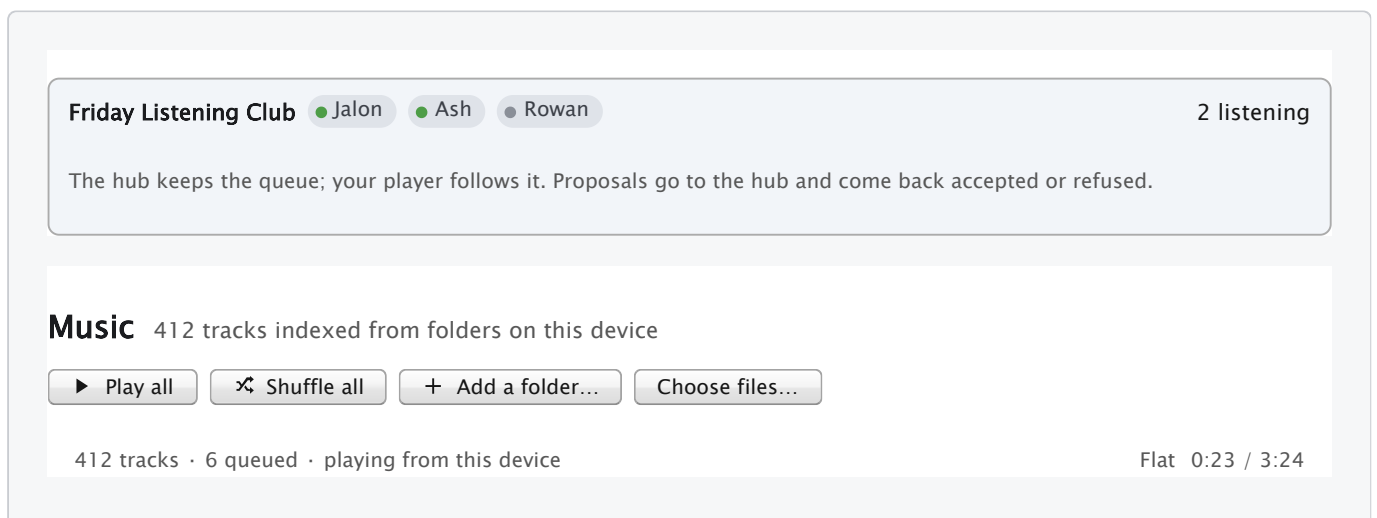
The desktop context menu with its playlist submenu; the New Playlist sheet; and the toast that answers a change. All three are the reference's, classes and all.

## THE EQUALIZER WINDOW



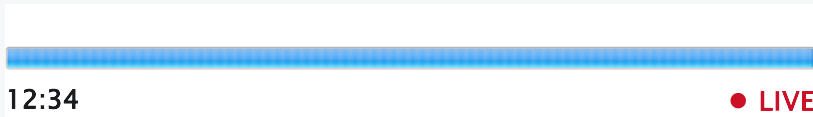
The iTunes equalizer: On beside the preset pop-up, then a preamp and the ten graphic centres on a  $\pm 12$  dB scale with tick dashes flanking each rail. The window hugs its rails at 600 px (DEC-014) rather than stretching to the page. A one-band preset is chosen here so the greyed rail is visible — those faders name the preset that is not using them.

## SHARED MODE, AND THE PAGE'S FURNITURE



The strip under the hero when you are in a group; a section's head and toolbar row; and the status line the bottom bar used to carry.

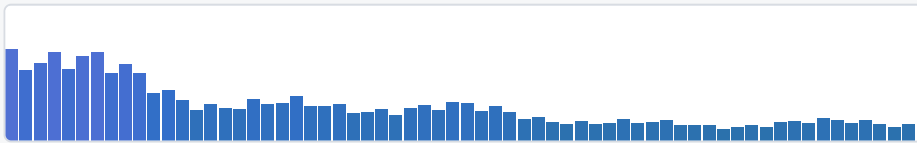
## A LIVE BROADCAST



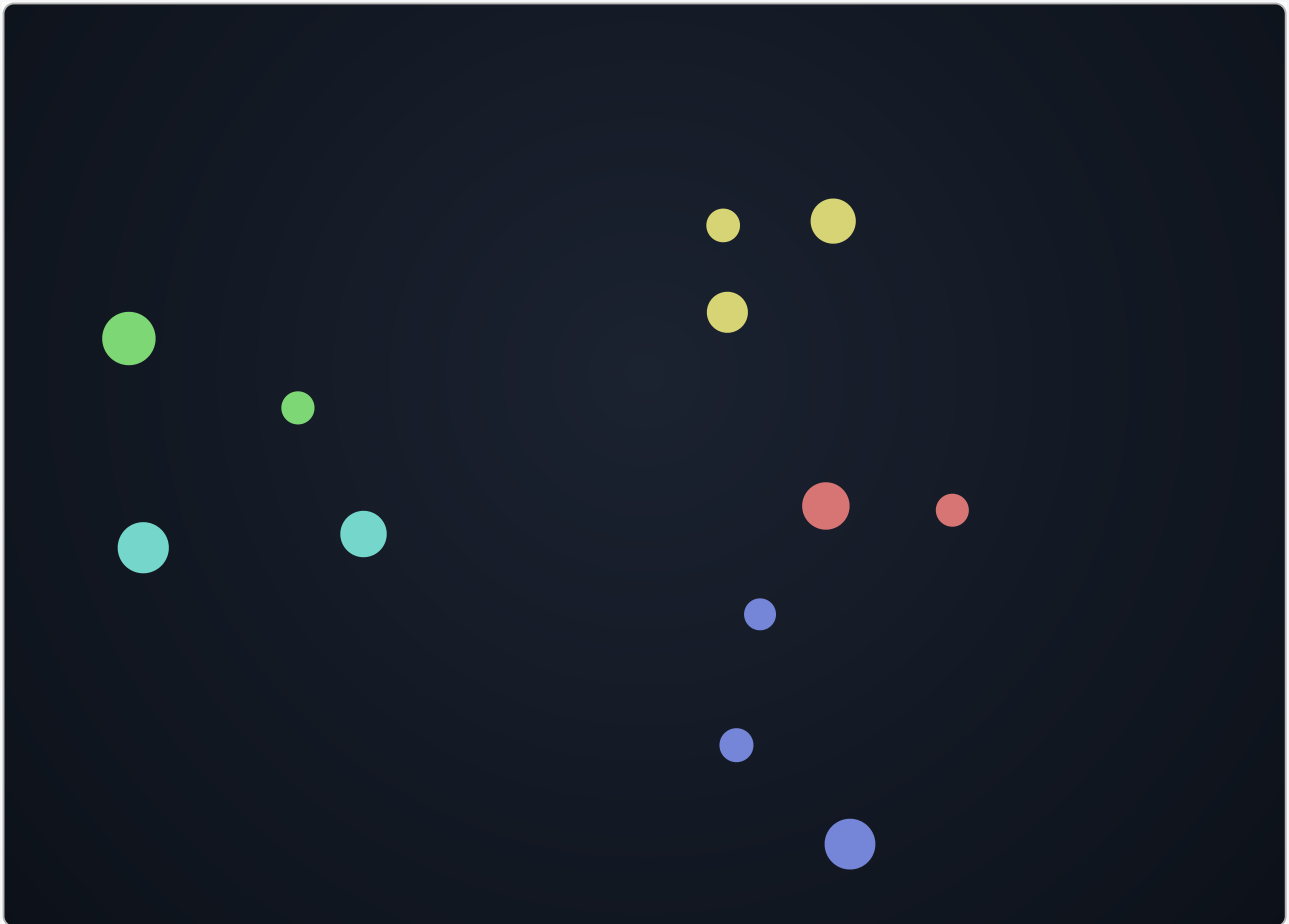
In shared mode the rail is not yours to drag: the duration stamp becomes the LIVE marker, the fill sits at the live edge, and the rail drops its slider role rather than announcing a control that refuses every change.

## THE VISUALISERS — SPECTRUM, STAGE AND CONSTELLATION

The player draws three visualisers, and all three follow one rule: a visualiser must carry the data, not hide it. The two below are drawn here by the player's own code — `drawSpectrum` and `layoutStars` in `music-player/src/lib/`, the functions the views call — in the player's own `.player-spectrum` and `.player-constellation` styles. The third, the **jewel case**, is live on the hero at the top of this chapter.



The spectrum, under “How this is being played” in the Now playing section. Each analyser frame is averaged into 64 bars; a bar's hue and lightness rise with its level, so a loud band reads brighter as well as taller. It draws only while a song plays — paused, the loop stops rather than repaint a still picture — and under reduced motion it redraws four times a second instead of every frame. There is no audio here, so the specimen is fed a generated frame; print shows one fixed frame.

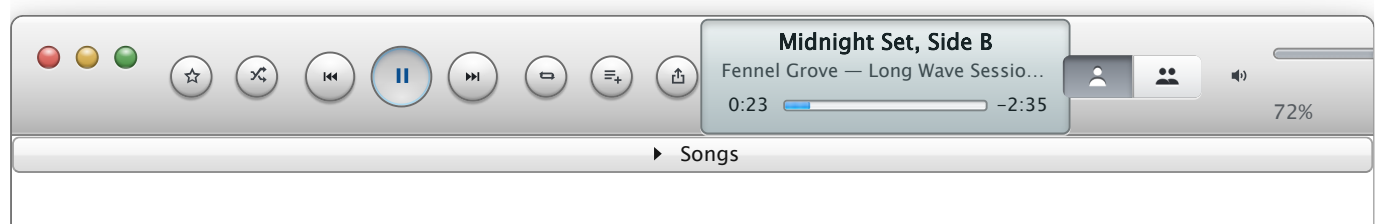


The constellation: the library as a star field, one star per album. Albums by one artist share an angular sector, a star's size is the album's length, and its colour is the artist's hue, so the layout is information, never scatter. The app renders it with WebGL and a slow rotation; this is the same placement projected through the same camera ( $55^\circ$ ,  $z = 42$ ), as SVG so it also prints. Hover a star for its album.

The **jewel case** is `packages/aqua-ui/src/stage/jewel-case.ts`: the case and disc built from primitives, sleeve and tray-card textures generated from the track's real cover and running order, a drag with momentum, and a pose remembered per device. It stands in for Cover Flow (DEC-006). The stage and the constellation load Three.js lazily and share the chunk, so a listener who opens neither never downloads it. All three keep their meaning without motion: the spectrum slows to a level, the stage stops its idle drift and keeps the CSS cover, and the constellation opens as the same albums in a table, with identical selection and keyboard behaviour (UX-MOTION-001, UX-STATE-001). Both sections also appear whole in the mockups: *Now playing (audio chain)* and *Constellation*.

## The window — what the hub and the companion are made of

The desktop skin: a framed window with traffic lights, a unified toolbar carrying the transport and the LCD, a source list, a work area with a table, and a bottom bar. The hub's admin GUI and the Windows companion are both built from this.



| #  | Name                       | Time | Artist                 | Album              | Genre      |
|----|----------------------------|------|------------------------|--------------------|------------|
| 1  | Track 1                    | 3:00 | Fennel Grove           | Long Wave Sessions | Ambient    |
| 2  | Track 2                    | 4:07 | Cassette Bloom         | Live from Pier 9   | Indie      |
| 3  | Track 3                    | 5:14 | Orbital Cartographers  | Copper Meridian    | Electronic |
| 4  | Ein außerordentlich langer | 6:21 | Marlow & the Tidewater | Quiet Arithmetic   | Folk       |
| 5  | Track 5                    | 3:28 | Fennel Grove           | Long Wave Sessions | Ambient    |
| 6  | Track 6                    | 4:35 | Cassette Bloom         | Live from Pier 9   | Indie      |
| 7  | Track 7                    | 5:42 | Orbital Cartographers  | Copper Meridian    | Electronic |
| 8  | Track 8                    | 6:49 | Marlow & the Tidewater | Quiet Arithmetic   | Folk       |
| 9  | Track 9                    | 3:56 | Fennel Grove           | Long Wave Sessions | Ambient    |
| 10 | Track 10                   | 4:03 | Cassette Bloom         | Live from Pier 9   | Indie      |
| 11 | Track 11                   | 5:10 | Orbital Cartographers  | Copper Meridian    | Electronic |
| 12 | Track 12                   | 6:17 | Marlow & the Tidewater | Quiet Arithmetic   | Folk       |
| 13 | Track 13                   | 3:24 | Fennel Grove           | Long Wave Sessions | Ambient    |
| 14 | Track 14                   | 4:31 | Cassette Bloom         | Live from Pier 9   | Indie      |
| 15 | Track 15                   | 5:38 | Orbital Cartographers  | Copper Meridian    | Electronic |
| 16 | Track 16                   | 6:45 | Marlow & the Tidewater | Quiet Arithmetic   | Folk       |
| 17 | Track 17                   | 3:52 | Fennel Grove           | Long Wave Sessions | Ambient    |
| 18 | Track 18                   | 4:59 | Cassette Bloom         | Live from Pier 9   | Indie      |
| 19 | Track 19                   | 5:06 | Orbital Cartographers  | Copper Meridian    | Electronic |
| 20 | Track 20                   | 6:13 | Marlow & the Tidewater | Quiet Arithmetic   | Folk       |
| 21 | Track 21                   | 3:20 | Fennel Grove           | Long Wave Sessions | Ambient    |

Every part of that window is a component: `AquaWindow`, `Toolbar`, `TrafficLights`, `Transport`, `LcdDisplay`, `Scrubber`, `VolumeSlider`, `SearchField`, `SegmentedControl`, `AvatarButton`, `SourceList`, `WorkArea`, `Content`, `AquaTable`, `Marquee`, `NowPlayingGlyph`, `BottomBar`, `IconButton`.

## Controls

Shared by both skins and all three products, verbatim.

Buttons: neutral / default / graphite / destructive / busy / disabled / small / mini / ellipsis / link

Cancel

Save

Graphite

⚠ Delete Playlist

Importing

Disabled

Small

Mini

Export...

Pressed

⬇ Download · 84 MB

Icon buttons: framed / plain / capsule / pressed / menu / disabled / large

☆

Checkbox / radio / text field / validation / pop-up

☒ Show explicit content

☒ Some folders

☐ Disabled

☒ Ask each time

☐ Windows only

Display name

Reference tuning

Disabled

Jalon

528

●

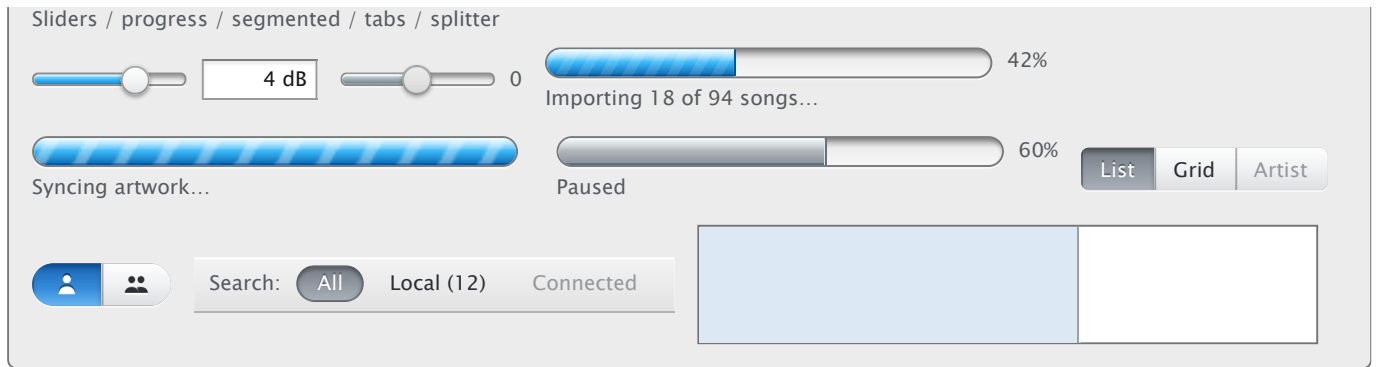
—

Shown to group members

● Reference must be between 400 and 480 Hz

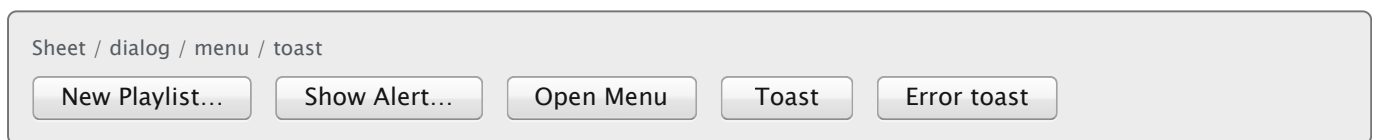
Default format

Original



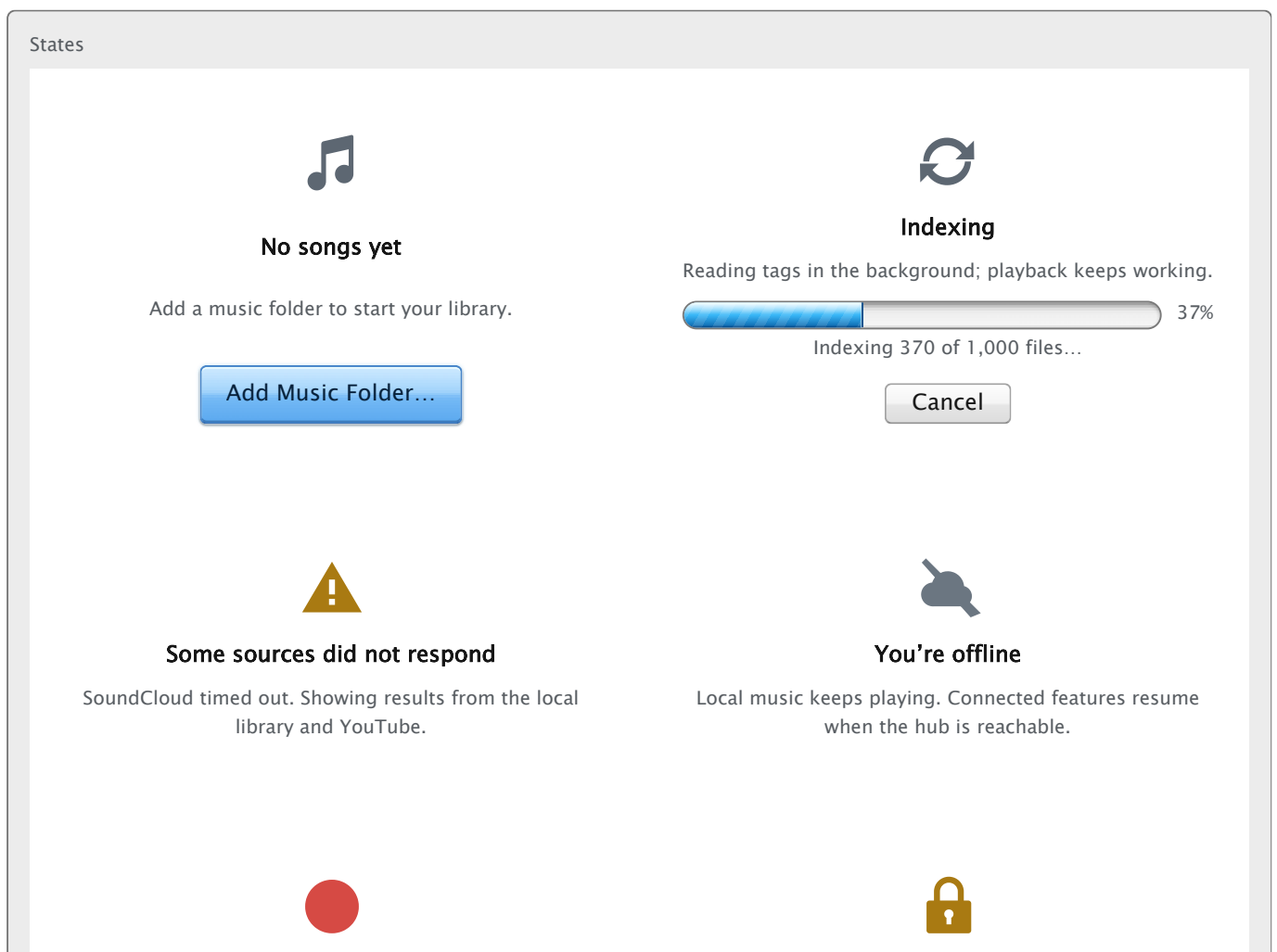
## Overlays

Sheets attach to their window; alerts stand alone; menus open where they are asked for and return focus to what asked. Press the buttons.



## States — the group people skip

Every screen owes an answer for empty, loading, offline, partial, refused and out-of-date. `UnavailableCapabilityState` exists so that “this cannot work here” is a designed state carrying its reason, rather than a blank pane.





Couldn't play this song

The file could not be decoded.

Skip

▼ Details

MEDIA\_ERR\_SRC\_NOT\_SUPPORTED — audio/x-ape is not decodable in this browser.



Upgrade required

This hub speaks protocol 3; this player supports up to 1.  
Nothing was changed.

Reconnect Folder

This browser needs permission again to read "Music".

Reconnect Folder

Remove



Download unavailable

YouTube does not permit audio downloads.

▼ Why unavailable?

The provider does not permit this — YouTube API Services Terms prohibit downloading audio.

Inline state + status dots + panel / form rows / list view



EQ unavailable for this source

Embedded YouTube audio is not exposed to Web Audio, so the equalizer is bypassed.

Audio

RETUNE

Reference tuning

440 Hz (standard)

Mode



Preserve tempo



Linked speed

Cents -31.77

Ratio 0.98182

Latency 43 ms

DEVICES

Studio PC

Windows companion · last seen 2 min ago



Kitchen

Player · online



Hub reachable



EQ unavailable



Refused

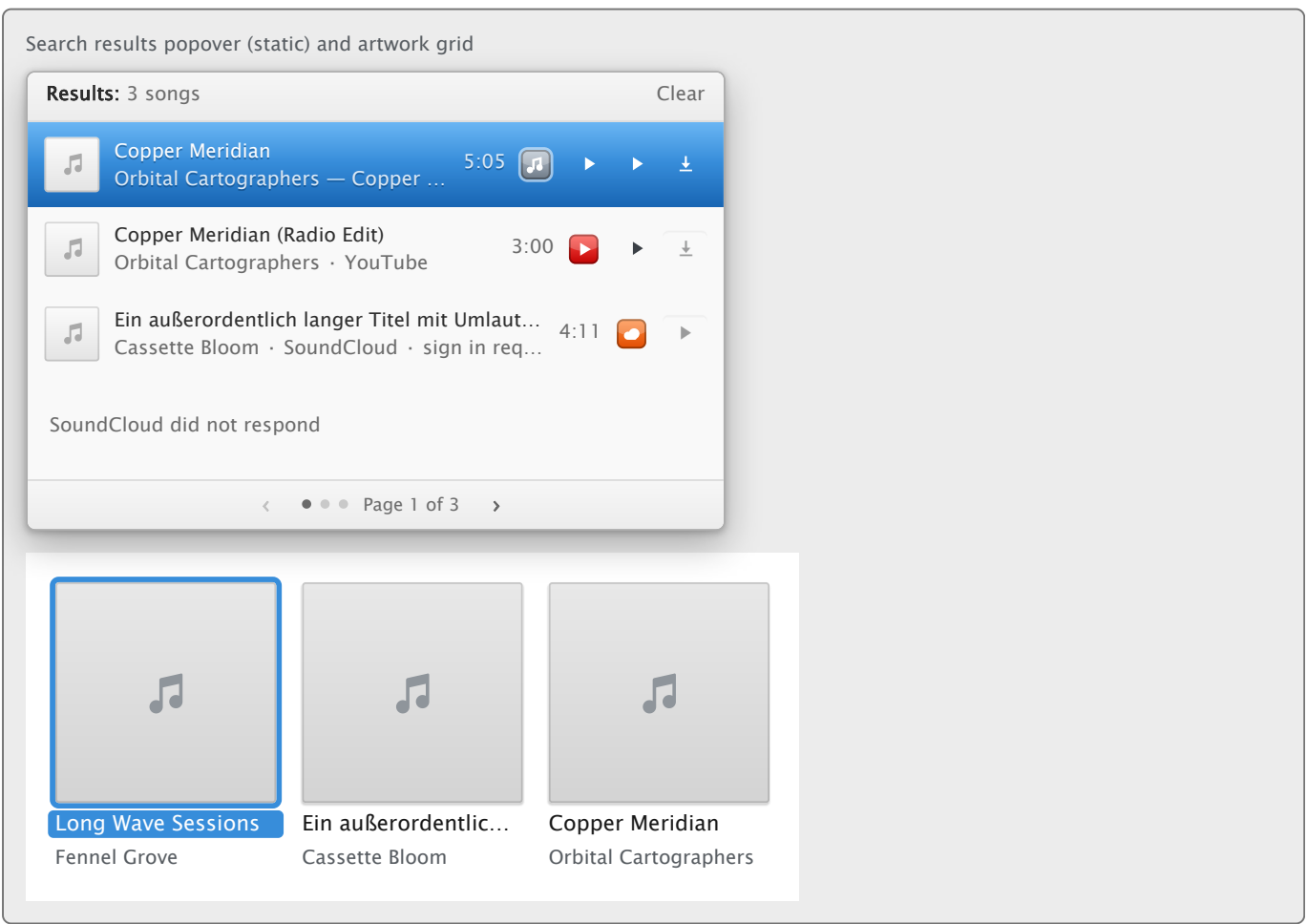


No

Status dots carry their text; colour is never the only signal.

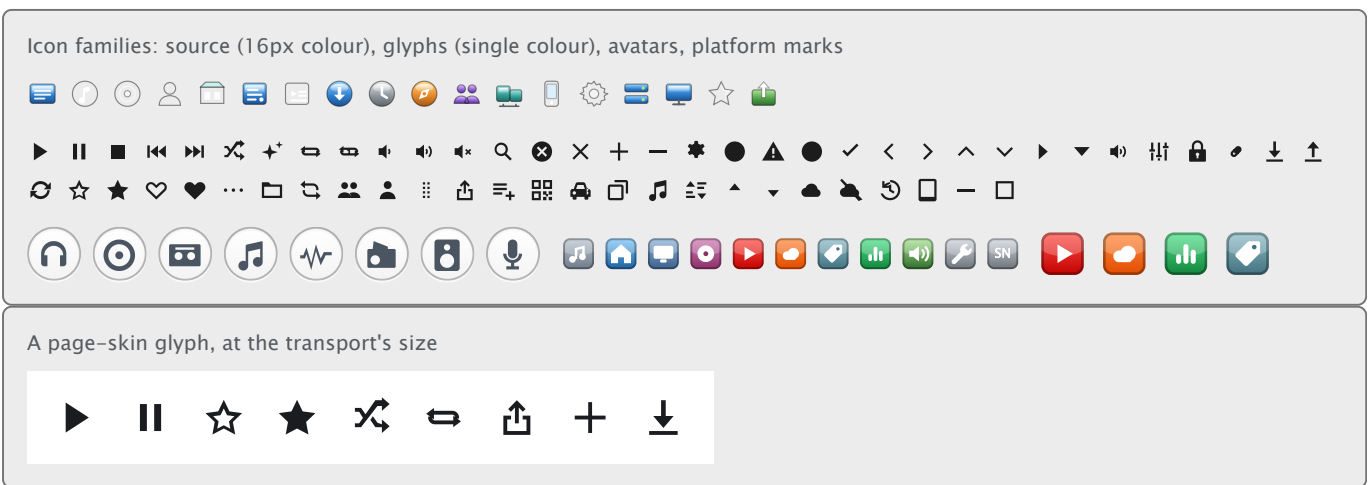
## Results and grid

The window skin's results popover — the same idea as the page's, at the toolbar's height — and the artwork grid.



## Icons

Three families: sixteen-pixel colour source icons for the sidebar, single-colour glyphs for everything else, and the avatar set.



## By product

Which element each product imports today, read from its source. The player is the page skin plus the shared controls; the hub and the companion are the window skin plus the same controls.

| ELEMENT          | PLAYER (PWA) | HUB ADMIN | COMPANION |
|------------------|--------------|-----------|-----------|
| PageBar          | •            | —         | —         |
| BarSearch        | •            | —         | —         |
| BarClock         | •            | —         | —         |
| ModeSwitch       | •            | —         | —         |
| ProfileButton    | •            | —         | —         |
| SectionStrip     | •            | —         | —         |
| Hero             | •            | —         | —         |
| HeroArt          | •            | —         | —         |
| JewelStage       | •            | —         | —         |
| TrackScrubber    | •            | —         | —         |
| KeyTransport     | •            | —         | —         |
| KeyButton        | •            | —         | —         |
| LevelSlider      | •            | —         | —         |
| MusicList        | •            | —         | —         |
| ButtonLink       | •            | —         | —         |
| IconButton       | •            | —         | —         |
| InlineValidation | •            | —         | —         |
| SegmentedControl | •            | —         | —         |
| Sheet            | •            | —         | —         |
| Slider           | •            | —         | —         |
| AquaTable        | •            | •         | •         |
| Button           | •            | •         | •         |
| Checkbox         | •            | •         | •         |
| EmptyState       | •            | •         | •         |
| Glyph            | •            | •         | •         |
| KeyValueList     | •            | •         | •         |
| Panel            | •            | •         | •         |
| PanelSection     | •            | •         | •         |
| PopUpMenu        | •            | •         | •         |
| ProgressBar      | •            | •         | •         |
|                  |              |           |           |

| ELEMENT       | PLAYER (PWA) | HUB ADMIN | COMPANION |
|---------------|--------------|-----------|-----------|
| SourceBadge   | •            | •         | •         |
| StatusDot     | •            | •         | •         |
| TextField     | •            | •         | •         |
| ToastProvider | •            | •         | •         |
| useToast      | •            | •         | •         |
| LoadingState  | •            | •         | —         |
| AquaWindow    | —            | •         | •         |
| Toolbar       | —            | •         | •         |
| SourceList    | —            | •         | •         |
| WorkArea      | —            | •         | •         |
| Content       | —            | •         | •         |
| BottomBar     | —            | •         | •         |
| SearchField   | —            | •         | •         |
| ErrorState    | —            | •         | —         |

The player also draws the search popover, the New Playlist sheet, the toast and the equalizer window with the page skin's classes (shown above) rather than through a component; their stylesheet block is the reference's own.

## Interaction and content rules

Every rule has a stable ID. Cite it in a change that touches the behaviour, and change the rule, the owning code and its test together. **Adopted** rules are the standard; **observed** ones describe what the code does today and have not been ratified. The source is `design/ux-rules.json`.

### PRINCIPLES THAT OUTRANK THE VISUALS

#### UX-PRIN-001 **A control that cannot act is not drawn**

● ADOPTED

No button that looks live and does nothing, no toggle whose state is decoration. If an action is impossible in this context the control is omitted or replaced by an explained state.

The equalizer window ships without traffic lights: three circles that close nothing would be decoration.

**OWNED BY** `music-player/src/views/Equalizer.tsx`

**CHECKED BY** reviewer Judgement call, checked in review (`docs/AQUA_CONFORMANCE.md`).  
`docs/DEVIATIONS.md`

#### UX-PRIN-002 **An unavailable capability says so where the choice is made**

● ADOPTED

Not in a tooltip, not in a log, not by disappearing. The control stays visible and, when chosen, reports the reason instead of acting.

The Shared segment of the mode switch stays visible with no hub paired and reports why when pressed.

**OWNED BY** `packages/aqua-ui/src/components/PageBar.tsx` `packages/aqua-ui/src/components/States.tsx`

**CHECKED BY** `packages/aqua-ui/tests/dom/page-chrome.test.tsx` — “reports the reason instead of switching when a mode is unavailable”  
`music-player/tests/e2e/player.spec.ts` — “the status bar carries the listening mode, and says why shared is unavailable”

#### UX-PRIN-003 **Say what the code does, not what the feature is called**

● ADOPTED

Labels describe the implemented behaviour. When a label and the DSP (or the network, or the provider) disagree, the label changes.

Solfeggio presets are described as filters with no claimed physical effect; retuning states that the fallback changes tempo.

**OWNED BY** `music-player/src/views/Equalizer.tsx` `music-player/src/views/Settings.tsx`

**CHECKED BY** `music-player/tests/e2e/player.spec.ts` — “the solfeggio presets are filters, and say so where the choice is made”  
`music-player/tests/e2e/player.spec.ts` — “retuning states how it is applied rather than claiming preserved tempo”

UX-PRIN-004 **A disabled control is still legible about why**

● ADOPTED

Greyed, not gone. The reason is part of the accessible name, not only a tooltip.

“500 Hz band, not used by 528 Hz (MI)”.

**OWNED BY** music-player/src/views/Equalizer.tsx

**CHECKED BY** music-player/tests/e2e/player.spec.ts — “the solfeggio presets are filters, and say so where the choice is made”

## KEYBOARD, FOCUS AND TOUCH

UX-KEY-001 **One tab stop per group; arrows move inside it**

● ADOPTED

Source lists, the section strip, the mode switch, tab strips, tables and menus use a roving tabIndex. Home/End jump, Enter activates, type-ahead selects. Focus travels with the selection.

**OWNED BY** packages/aqua-ui/src/components/SourceList.tsx packages/aqua-ui/src/components/SectionStrip.tsx  
packages/aqua-ui/src/components/AquaTable.tsx packages/aqua-ui/src/components/MusicList.tsx  
docker-container/src/web/App.tsx

**CHECKED BY** docker-container/tests/e2e/a11y.spec.ts — “@a11y the tab strip is navigable with the keyboard alone”  
packages/aqua-ui/tests/dom/source-list.test.tsx — “has one tab stop, moves with arrows, Home/End, selects with Enter and supports type-ahead”  
packages/aqua-ui/tests/dom/table.test.tsx — “uses a roving tabIndex, arrow navigation, Space selection and Enter activation”  
packages/aqua-ui/tests/dom/page-chrome.test.tsx — “has one tab stop, moves with arrows and Home/End, chooses with Enter, and answers type-ahead”

UX-FOCUS-001 **Focus is always visible**

● ADOPTED

Every focusable element shows --aqua-focus-shadow or a 2px --np-accent outline at 2px offset, including when the ring is drawn on an ancestor via :focus-within.

**OWNED BY** packages/aqua-ui/src/styles/aqua.css packages/aqua-ui/src/styles/now-playing.css

**CHECKED BY** music-player/tests/e2e/a11y.spec.ts — “@a11y focus is visible wherever it lands”

UX-KEY-002 **Every action is reachable with the keyboard alone**

● ADOPTED

Every section and every transport control can be reached and operated without a pointer. Space toggles playback without scrolling the page.

**OWNED BY** music-player/src/App.tsx docker-container/src/web/App.tsx

**CHECKED BY** music-player/tests/e2e/a11y.spec.ts — “@a11y every section is reachable with the keyboard alone”  
music-player/tests/e2e/player.spec.ts — “space toggles playback and does not scroll the page”  
docker-container/tests/e2e/a11y.spec.ts — “@a11y the tab strip is navigable with the keyboard alone”

#### UX-TOUCH-001 **Nothing interactive is under 44px on a coarse pointer**

● ADOPTED

Under (pointer: coarse) controls grow; where the visual cannot grow, a transparent `::after` takes the taps. Desktop keeps its 18px rows. Hit targets are never shrunk to match a visual; `--aqua-hit` is 32px on desktop, 44px on touch.

**OWNED BY** `packages/aqua-ui/src/styles/aqua.css` `packages/aqua-ui/src/styles/now-playing.css`

**CHECKED BY** `music-player/tests/e2e/responsive.spec.ts` — “nothing overflows, no text under 12px, no target under 44px”

`music-player/tests/e2e/responsive.spec.ts` — “the desktop keeps the design it was drawn with”

`packages/aqua-ui/tests/e2e/styleguide.spec.ts` — “every screen of every product fits at every size”

#### UX-TOUCH-002 **No text under 12px on touch; field text is 16px**

● ADOPTED

The touch layer steps the scale up (15/14/13/12/11px). A field's text is 16px because iOS zooms the page when a smaller field takes focus.

**OWNED BY** `packages/aqua-ui/src/styles/aqua.css` `packages/aqua-ui/src/styles/now-playing.css`

**CHECKED BY** `music-player/tests/e2e/responsive.spec.ts` — “settings, where every panel of explanatory text lives”

### STATES, FEEDBACK AND MOTION

#### UX-STATE-001 **Every data view answers empty, loading and error**

● ADOPTED

A view that fetches renders `LoadingState` while waiting, `ErrorState` with a retry when it fails, and an `EmptyState` that says what to do next. Blank panes are defects.

**OWNED BY** `docker-container/src/web/views/common.tsx` `docker-container/src/web/ViewBoundary.tsx`  
`packages/aqua-ui/src/components/States.tsx`

**CHECKED BY** `music-player/tests/e2e/a11y.spec.ts` — “@a11y the empty library state is announced as a group with a name”

`music-player/tests/e2e/player.spec.ts` — “loads with no console errors and an honest empty state”

#### UX-MOTION-001 **Motion confirms; reduced motion is a contract**

● ADOPTED

Nothing animates that a person did not just cause. Under `prefers-reduced-motion` the live dot stops, marquees hold their ellipsis, sheets fade without travel, and the 3D stage stops drifting. New animations read `--aqua-anim-state` or `--aqua-sheet-travel`.

**OWNED BY** `packages/aqua-ui/src/styles/aqua.css` `packages/aqua-ui/src/styles/now-playing.css`  
`packages/aqua-ui/src/components/Marquee.tsx`

**CHECKED BY** `music-player/tests/e2e/a11y.spec.ts` — “@a11y respects reduced motion”

`packages/aqua-ui/tests/unit/aqua-conformance.test.ts` — “§17.5”

#### UX-FEED-001 **Progress says what is happening in words**

● ADOPTED

Progress bars carry a name and a value; an indeterminate bar is `aria-busy` and has text. Live regions are polite and carry one sentence.

**OWNED BY** `packages/aqua-ui/src/components/ProgressBar.tsx` `packages/aqua-ui/src/components/Toast.tsx`

**CHECKED BY** `packages/aqua-ui/tests/dom/controls.test.tsx` — “progress has a name and value; indeterminate is busy with text”

UX-A11Y-001 **Roles are true and colour is never the only signal**

● ADOPTED

A rail that cannot be dragged is role=img, not a slider that refuses changes. Icon buttons have names. The playing row has a glyph as well as a tint; status dots carry text.

**OWNED BY** packages/aqua-ui/src/components/Scrubber.tsx packages/aqua-ui/src/components/IconButton.tsx  
packages/aqua-ui/src/components/Badge.tsx

**CHECKED BY** packages/aqua-ui/tests/dom/controls.test.tsx — “live scrubber is not a slider and shows the LIVE marker”  
packages/aqua-ui/tests/dom/controls.test.tsx — “icon buttons require names; menu affordance sets aria-haspopup”

UX-LINK-001 **A download is a link**

● ADOPTED

Downloads and destinations in another tab use ButtonLink (an <a> wearing the button face), never a <button> that assigns location.

**OWNED BY** packages/aqua-ui/src/components/Button.tsx

**CHECKED BY** music-player/tests/e2e/player.spec.ts — “Settings opens from the avatar and carries the equalizer and the companion download”

## TRUST, SAFETY AND DESTRUCTIVE ACTIONS

UX-SAFE-001 **Destructive actions are worded plainly and confirmed**

● ADOPTED

The button names the consequence (“Forget this hub”, “Remove folder”), sits apart from the default action, and the hub and companion ask before doing it.

**OWNED BY** packages/aqua-ui/src/components/Sheet.tsx docker-container/src/web/views/Devices.tsx  
windows-companion/src/renderer/views/Folders.tsx

**CHECKED BY** packages/aqua-ui/tests/dom/controls.test.tsx  
docs/AQUA\_CONFORMANCE.md  
reviewer Each new destructive action is reviewed for wording and confirmation.

UX-SAFE-002 **The bootstrap password must be replaced before anything else**

● ADOPTED

admin/admin signs in once, straight into Change password. Every setup-gated screen refuses until then, and the hub stays on loopback outside a container.

**OWNED BY** docker-container/src/web/App.tsx docker-container/src/auth/service.ts

**CHECKED BY** docker-container/tests/integration/setup-and-pairing.test.ts — “refuses every setup-gated route until the password is changed”  
docker-container/tests/e2e/a11y.spec.ts — “@a11y the sign-in screen has no detectable violations”

UX-SAFE-003 **Pairing is confirmed by comparing a fingerprint**

● ADOPTED

A device receives a scoped credential only after the administrator confirms the fingerprint shown on both screens. Codes are never listed again after they are issued.

**OWNED BY** docker-container/src/web/views/Devices.tsx docker-container/src/pairing/service.ts

**CHECKED BY** docker-container/tests/integration/setup-and-pairing.test.ts — “issues a scoped credential only after the fingerprint is confirmed”



UX-SAFE-004 **Privacy claims are stated where they can be checked**

● ADOPTED

Copy that says nothing is uploaded, copied or moved must be true of the code path beside it, and the player loads nothing from outside its own origin.

**OWNED BY** music-player/src/views/Settings.tsx music-player/src/views/Metrics.tsx

**CHECKED BY** music-player/tests/e2e/player.spec.ts — “privacy claims are stated where a person can check them”  
music-player/tests/e2e/player.spec.ts — “loads nothing from outside its own origin”

## SHARED LISTENING AND THE DISCORD BOT

UX-SHARED-001 **The shared queue rejects stale commands and says so**

● ADOPTED

Every queue command carries the revision it was based on. A stale command is refused with the current revision, never applied over someone else's change.

**OWNED BY** docker-container/src/group/service.ts packages/domain/src/queue.ts

**CHECKED BY** docker-container/tests/integration/groups-and-queue.test.ts

UX-DISCORD-001 **Slash and prefix commands behave identically**

● ADOPTED

Both transports run through one CommandService and render the same templates; the only difference is how the reply is delivered.

**OWNED BY** docker-container/src/group/command-service.ts docker-container/src/discord/gateway.ts

**CHECKED BY** docker-container/tests/integration/discord-parity.test.ts — “produces identical outcomes for every command”

UX-DISCORD-002 **Refusals are private and say what was needed**

● ADOPTED

permissionDenied, noResults, unavailableSource, emptyQueue, wrongChannel and error replies are ephemeral and name the reason or the channel to use.

**OWNED BY** packages/domain/src/templates.ts docker-container/src/group/command-service.ts

**CHECKED BY** docker-container/tests/integration/discord-parity.test.ts — “applies the same permission decision on both transports”

UX-DISCORD-003 **The bot never goes silent without a reason**

● ADOPTED

A track the bot cannot play is skipped with the reason recorded; a failed decode advances as an error; the admin panel shows the voice state and last error.

**OWNED BY** docker-container/src/discord/voice.ts docker-container/src/discord/worker.ts

**CHECKED BY** docker-container/tests/unit/discord-voice.test.ts — “skips a track it cannot play instead of going silent”  
docker-container/tests/integration/discord-worker.test.ts — “shows the worker's live status in the hub”

## WORDS

#### VOICE-001 Empty states say what to do next

● OBSERVED

Title names the situation in a few plain words ("Nothing queued"); text gives the next action or the reason. No apologies, no exclamation marks.

"Nothing queued — Play something from your library and it will appear here."

**OWNED BY** music-player/src/views/Queue.tsx windows-companion/src/renderer/views/Transfers.tsx

**CHECKED BY** music-player/src/views/Queue.tsx

#### VOICE-002 Errors explain the problem and the way out

● ADOPTED

Name what failed in the person's terms and what fixes it ("Add a bot token first (Admin → Discord → Token)"). Never show a stack trace or an internal identifier; a 500 carries a correlation id instead.

**OWNED BY** docker-container/src/app.ts docker-container/src/discord/service-interface.ts

**CHECKED BY** docker-container/src/discord/service-interface.ts

#### VOICE-003 One product name, set in one place

● ADOPTED

"Now Playing" and the product names come from BRANDING in TypeScript. Native resources that cannot import it (Android strings.xml, the SVG icons' aria-labels, the Electron builder config) repeat it and must be changed together.

**OWNED BY** packages/contracts/src/branding.ts android/app/src/main/res/values/strings.xml  
music-player/public/icon.svg windows-companion/resources/icon.svg

**CHECKED BY** packages/contracts/tests/branding.test.ts  
packages/contracts/src/branding.ts

## Accessibility

Not bolted on afterwards. These are parts of the design.

- **One tab stop per group, arrows to move.** The section strip, the mode switch, the music list and the context menu all use a roving tabIndex, and the focus ring travels with the selection.
- **Focus is always visible.** An end-to-end test tabs the whole page and fails on any element that shows no indicator.
- **Roles must be true.** A rail that cannot be dragged is role="img" with no value pair, not a slider that refuses every change. A button that opens a section does not claim aria-haspopup="menu".
- **Names say what will happen**, including the refusal.
- **Colour is never the only signal.** The playing row has a glyph as well as a tint; status dots carry text.

## Every screen

Every surface found in the products' own navigation, with the states it has to answer and the rules that govern it. `pnpm styleguide:check` rebuilds this list from the `ViewId` declarations, the sheet files and the Discord command table, and fails when something appears that is not recorded in `design/coverage.json`.

**Seen running:** opened in the running product for this revision (see Coverage). **Mockup only:** rendered at every size in the mockups from the real components with fixture data. **Not run here:** the product could not be started where this guide was built. **Source only:** checked against the code, not looked at.

### MUSIC PLAYER

Page skin. The PWA, the single-file build, the Android app and the local helper all show this.

| SURFACE                                                                                                                                                                      | STATES IT MUST ANSWER                                                                                                                                                                                                                                                                                                                                                                                     | RULES                                                   | SEEN           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|----------------|
| <b>Status bar, mode switch, hero and section strip</b><br>every section<br><code>music-player/src/App.tsx</code><br><code>packages/aqua-ui/src/components/PageBar.tsx</code> | <ul style="list-style-type: none"><li>nothing playing</li><li>playing</li><li>paused</li><li>live stream (read-only scrubber)</li><li>unknown duration</li><li>shared unavailable (reason shown)</li><li>narrow (strip scrolls)</li><li>transport: volume beneath when it cannot flank the keys (DEC-016)</li><li>transport: keys wrapped, previous/play/next alone on the first line (DEC-016)</li></ul> | UX-PRIN-002<br>UX-KEY-001<br>UX-KEY-002<br>UX-TOUCH-001 | ● seen running |
| <b>Music Library</b><br>section strip → Music Library<br><code>music-player/src/views/Library.tsx</code><br><code>packages/aqua-ui/src/components/MusicList.tsx</code>       | <ul style="list-style-type: none"><li>empty (how to add music)</li><li>scanning</li><li>loaded</li><li>sorted</li><li>selected row</li><li>playing row</li><li>unplayable format</li></ul>                                                                                                                                                                                                                | UX-KEY-001<br>UX-STATE-001<br>UX-A11Y-001<br>VOICE-001  | ● seen running |
| <b>Now playing</b><br>section strip → Now playing<br><code>music-player/src/views/NowPlaying.tsx</code><br><code>packages/domain/src/visualisers/spectrum.ts</code>          | <ul style="list-style-type: none"><li>nothing playing</li><li>playing (spectrum moving)</li><li>paused (spectrum stopped)</li><li>reduced motion (spectrum at 4 per second)</li><li>equalizer not applied (reason shown)</li><li>shared (group queue)</li></ul>                                                                                                                                           | UX-STATE-001<br>UX-MOTION-001                           | ○ mockup only  |
| <b>Queue</b><br>section strip → Queue<br><code>music-player/src/views/Queue.tsx</code>                                                                                       | <ul style="list-style-type: none"><li>nothing queued</li><li>queued</li><li>shuffled</li><li>shared (hub revision)</li></ul>                                                                                                                                                                                                                                                                              | UX-STATE-001<br>UX-SHARED-001<br>VOICE-001              | ● seen running |
| <b>Playlists</b><br>section strip → Playlists<br><code>music-player/src/views/Playlists.tsx</code>                                                                           | <ul style="list-style-type: none"><li>no playlists</li><li>list</li><li>playlist open</li><li>empty playlist</li></ul>                                                                                                                                                                                                                                                                                    | UX-STATE-001<br>VOICE-001                               | ○ source only  |

| SURFACE                                                                                                                                                                                  | STATES IT MUST ANSWER                                                                                                                                                                                                                          | RULES                                                                                   | SEEN           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------|
| <b>Search</b><br>status-bar search → Enter<br>music-player/src/views/Search.tsx<br>music-player/src/components/SearchPopover.tsx                                                         | <ul style="list-style-type: none"> <li>prompt (device only / with hub)</li> <li>searching hub</li> <li>no local matches</li> <li>results</li> <li>hub unreachable</li> </ul>                                                                   | UX-STATE-001<br>UX-PRIN-002                                                             | ○ source only  |
| <b>Listening history</b><br>section strip → Listening history<br>music-player/src/views/Metrics.tsx                                                                                      | <ul style="list-style-type: none"> <li>nothing yet</li> <li>history</li> </ul>                                                                                                                                                                 | UX-STATE-001<br>UX-SAFE-004                                                             | ● seen running |
| <b>Constellation</b><br>section strip → Constellation<br>music-player/src/views/Constellation.tsx<br>packages/domain/src/visualisers/constellation-layout.ts                             | <ul style="list-style-type: none"> <li>nothing to map</li> <li>field</li> <li>table (equivalent view)</li> <li>reduced motion (table)</li> <li>WebGL unavailable</li> </ul>                                                                    | UX-MOTION-001<br>UX-STATE-001                                                           | ○ mockup only  |
| <b>Settings and equalizer</b><br>profile button in the status bar<br>music-player/src/views/Settings.tsx<br>music-player/src/views/Equalizer.tsx                                         | <ul style="list-style-type: none"> <li>no folders</li> <li>folders</li> <li>hub paired</li> <li>hub not paired</li> <li>equalizer on/off</li> <li>sofeggio preset (dimmed bands)</li> <li>crossfade</li> <li>car/lock-screen report</li> </ul> | UX-PRIN-001<br>UX-PRIN-003<br>UX-PRIN-004<br>UX-LINK-001<br>UX-TOUCH-002<br>UX-SAFE-004 | ● seen running |
| <b>Add to playlist</b><br>+ beside the transport, or row menu<br>music-player/src/components/AddToPlaylistSheet.tsx                                                                      | <ul style="list-style-type: none"> <li>open</li> <li>no playlists</li> </ul>                                                                                                                                                                   | UX-KEY-001                                                                              | ○ source only  |
| <b>New playlist</b><br>Playlists → New, or row menu<br>music-player/src/components/NewPlaylistSheet.tsx<br>packages/aqua-ui/src/components/music-list-behaviours.tsx                     | <ul style="list-style-type: none"> <li>empty name (validation)</li> <li>valid</li> </ul>                                                                                                                                                       | VOICE-002                                                                               | ○ source only  |
| <b>Download from the hub</b><br>row menu → Download<br>music-player/src/components/DownloadSheet.tsx                                                                                     | <ul style="list-style-type: none"> <li>choose format</li> <li>basis required</li> <li>in progress</li> <li>failed with reason</li> </ul>                                                                                                       | UX-FEED-001<br>UX-PRIN-002                                                              | ○ source only  |
| <b>Fetch with a local tool</b><br>Library → Fetch (when a helper or the Android tools are present)<br>music-player/src/components/FetchSheet.tsx<br>music-player/src/lib/fetch-helper.ts | <ul style="list-style-type: none"> <li>no tool (control hidden)</li> <li>url entry</li> <li>running</li> <li>finished (files)</li> <li>failed (tool's own reason)</li> </ul>                                                                   | UX-PRIN-001<br>UX-FEED-001<br>VOICE-002                                                 | ○ source only  |
|                                                                                                                                                                                          |                                                                                                                                                                                                                                                |                                                                                         |                |

| SURFACE                                                                                                                                                | STATES IT MUST ANSWER                                                                                                            | RULES                     | SEEN           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------|
| <b>Share</b><br>share button beside the transport, or a playlist<br>music-player/src/components/ShareSheet.tsx                                         | <ul style="list-style-type: none"> <li>no hub (explains why)</li> <li>link created</li> <li>copied</li> </ul>                    | UX-PRIN-002               | ○ source only  |
| <b>Search popover</b><br>typing in the status-bar search<br>music-player/src/components/SearchPopover.tsx                                              | <ul style="list-style-type: none"> <li>counting</li> <li>results</li> <li>none</li> <li>dismissed (arrow brings back)</li> </ul> | UX-FEED-001<br>UX-KEY-001 | ● seen running |
| <b>Notice bar and toasts</b><br>any warning the store raises<br>music-player/src/components/NoticeBar.tsx<br>packages/aqua-ui/src/components/Toast.tsx | <ul style="list-style-type: none"> <li>info</li> <li>warning</li> <li>dismissed</li> </ul>                                       | UX-FEED-001<br>VOICE-002  | ● seen running |
| <b>Row context menu</b><br>right-click or the row's menu key<br>packages/aqua-ui/src/components/music-list-behaviours.tsx                              | <ul style="list-style-type: none"> <li>open</li> <li>disabled items with reason</li> </ul>                                       | UX-KEY-001<br>UX-PRIN-004 | ○ source only  |

## HUB ADMIN AND PUBLIC PAGES

Window skin, served by the container on port 4546.

| SURFACE                                                                                                                     | STATES IT MUST ANSWER                                                                                                                                                                            | RULES                                      | SEEN           |
|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|----------------|
| <b>Sign in</b><br>any admin URL without a session<br>docker-container/src/web/App.tsx                                       | <ul style="list-style-type: none"> <li>first run (admin/admin hint)</li> <li>wrong password</li> <li>rate limited</li> </ul>                                                                     | UX-SAFE-002<br>VOICE-002                   | ● seen running |
| <b>Change the bootstrap password</b><br>first sign-in<br>docker-container/src/web/App.tsx                                   | <ul style="list-style-type: none"> <li>weak password refused</li> <li>saved</li> </ul>                                                                                                           | UX-SAFE-002                                | ○ source only  |
| <b>View error boundary</b><br>any admin view that throws<br>docker-container/src/web/ViewBoundary.tsx                       | <ul style="list-style-type: none"> <li>error with retry</li> </ul>                                                                                                                               | UX-STATE-001                               | ○ source only  |
| <b>Overview</b><br>Overview tab<br>docker-container/src/web/views/Overview.tsx<br>docker-container/src/web/views/common.tsx | <ul style="list-style-type: none"> <li>loading</li> <li>loaded (tiles: hub, connections, groups, providers, storage, jobs)</li> <li>error</li> <li>alerts (count on the Overview tab)</li> </ul> | UX-STATE-001<br>UX-KEY-001                 | ○ mockup only  |
| <b>Devices and pairing</b><br>Devices tab → Devices<br>docker-container/src/web/views/Devices.tsx                           | <ul style="list-style-type: none"> <li>no devices</li> <li>pairing code shown</li> <li>claim waiting for fingerprint</li> <li>revoke (confirm)</li> </ul>                                        | UX-SAFE-001<br>UX-SAFE-003<br>UX-STATE-001 | ○ source only  |
|                                                                                                                             |                                                                                                                                                                                                  |                                            |                |

| SURFACE                                                                                                                                        | STATES IT MUST ANSWER                                                                                                                                                                                           | RULES                                      | SEEN          |
|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|---------------|
| <b>Groups</b><br>Groups tab<br>docker-<br>container/src/web/views/Groups.tsx                                                                   | <ul style="list-style-type: none"> <li>no groups</li> <li>group with queue</li> <li>archived</li> <li>new group</li> <li>invite made (code shown once)</li> <li>invite withdrawn (confirm)</li> </ul>           | UX-SHARED-001<br>UX-STATE-001              | ○ source only |
| <b>Providers</b><br>Music tab → Providers<br>docker-<br>container/src/web/views/Providers.tsx                                                  | <ul style="list-style-type: none"> <li>available</li> <li>sign in needed</li> <li>not configured</li> <li>off</li> <li>test result</li> </ul>                                                                   | UX-PRIN-002<br>UX-A11Y-001                 | ○ mockup only |
| <b>Library</b><br>Music tab → Library<br>docker-<br>container/src/web/views/Library.tsx                                                        | <ul style="list-style-type: none"> <li>no roots</li> <li>scanning</li> <li>tracks</li> </ul>                                                                                                                    | UX-STATE-001                               | ○ source only |
| <b>Downloads</b><br>Music tab → Downloads<br>docker-<br>container/src/web/views/Downloads.tsx                                                  | <ul style="list-style-type: none"> <li>queued</li> <li>running</li> <li>paused</li> <li>failed with reason</li> <li>completed</li> <li>formats unavailable without FFmpeg</li> </ul>                            | UX-FEED-001<br>UX-PRIN-002                 | ○ mockup only |
| <b>Profiles</b><br>Devices tab → Profiles<br>docker-<br>container/src/web/views/Profiles.tsx                                                   | <ul style="list-style-type: none"> <li>none</li> <li>listed</li> <li>renaming</li> <li>name taken</li> <li>remove picture (confirm)</li> </ul>                                                                  | UX-SAFE-001<br>UX-FEED-001                 | ○ source only |
| <b>Shared links</b><br>Sharing tab → Shared links<br>docker-<br>container/src/web/views/Shares.tsx                                             | <ul style="list-style-type: none"> <li>none</li> <li>active</li> <li>revoked (confirm)</li> </ul>                                                                                                               | UX-SAFE-001                                | ○ source only |
| <b>Recommendations</b><br>Music tab → Recommendations<br>docker-<br>container/src/web/views/Recommendations.tsx                                | <ul style="list-style-type: none"> <li>defaults</li> <li>edited</li> </ul>                                                                                                                                      | UX-PRIN-003                                | ○ source only |
| <b>Discord</b><br>Sharing tab → Discord<br>docker-<br>container/src/web/views/Discord.tsx<br>docker-container/src/discord/service-interface.ts | <ul style="list-style-type: none"> <li>no token</li> <li>configured but disabled</li> <li>worker not running</li> <li>connected</li> <li>voice playing</li> <li>last error</li> <li>template preview</li> </ul> | UX-PRIN-002<br>UX-DISCORD-003<br>VOICE-002 | ○ source only |
| <b>Network</b><br>System tab → Network<br>docker-<br>container/src/web/views/Network.tsx                                                       | <ul style="list-style-type: none"> <li>localhost</li> <li>lan</li> <li>remote (restart required)</li> </ul>                                                                                                     | UX-SAFE-002                                | ○ source only |

| SURFACE                                                                                              | STATES IT MUST ANSWER                                                                                                                                                                                                      | RULES                       | SEEN          |
|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------|
| <b>Backup</b><br>System tab → Backup<br>docker-<br>container/src/web/views/Backup.tsx                | <ul style="list-style-type: none"> <li>no backups</li> <li>list</li> <li>restore (confirm)</li> <li>restore failed with reason</li> <li>space measured</li> <li>space unknown</li> <li>next backup will not fit</li> </ul> | UX-SAFE-001<br>UX-STATE-001 | ○ source only |
| <b>Diagnostics</b><br>System tab → Diagnostics<br>docker-<br>container/src/web/views/Diagnostics.tsx | <ul style="list-style-type: none"> <li>loaded</li> </ul>                                                                                                                                                                   | UX-STATE-001                | ○ source only |
| <b>Public share page</b><br>/s/<token><br>docker-<br>container/src/api/routes/shares.ts<br>DEC-008   | <ul style="list-style-type: none"> <li>track or playlist</li> <li>stream allowed</li> <li>link unavailable</li> </ul>                                                                                                      | UX-SAFE-004                 | ○ source only |

## WINDOWS COMPANION

Window skin in an Electron window.

| SURFACE                                                                                                      | STATES IT MUST ANSWER                                                                                                                                                | RULES                      | SEEN           |
|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------|
| <b>Folders</b><br>source list → Library → Folders<br>windows-<br>companion/src/renderer/views/Folders.tsx    | <ul style="list-style-type: none"> <li>no folders</li> <li>watching</li> <li>drive not connected</li> <li>remove (confirm)</li> </ul>                                | UX-SAFE-001<br>VOICE-001   | ○ not run here |
| <b>Music</b><br>source list → Library → Music<br>windows-<br>companion/src/renderer/views/Library.tsx        | <ul style="list-style-type: none"> <li>nothing indexed</li> <li>no matches</li> <li>tracks</li> </ul>                                                                | UX-STATE-001<br>VOICE-001  | ○ not run here |
| <b>Connection</b><br>source list → Hub → Connection<br>windows-<br>companion/src/renderer/views/Hub.tsx      | <ul style="list-style-type: none"> <li>not paired</li> <li>pairing (fingerprint)</li> <li>connected</li> <li>plain-http warning</li> <li>forget (confirm)</li> </ul> | UX-SAFE-001<br>UX-SAFE-003 | ○ not run here |
| <b>Transfers</b><br>source list → Hub → Transfers<br>windows-<br>companion/src/renderer/views/Transfers.tsx  | <ul style="list-style-type: none"> <li>nothing transferring (paired / not paired)</li> <li>uploading</li> <li>cancelled</li> <li>failed</li> </ul>                   | UX-FEED-001<br>VOICE-001   | ○ not run here |
| <b>Backup</b><br>source list → This computer → Backup<br>windows-<br>companion/src/renderer/views/Backup.tsx | <ul style="list-style-type: none"> <li>export</li> <li>import (validated)</li> <li>invalid backup refused</li> </ul>                                                 | UX-SAFE-001<br>VOICE-002   | ○ not run here |
|                                                                                                              |                                                                                                                                                                      |                            |                |

| SURFACE                                                                                               | STATES IT MUST ANSWER                                                   | RULES     | SEEN                               |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------|------------------------------------|
| <b>About</b><br>source list → This computer → About<br>windows-companion/src/renderer/views/About.tsx | <ul style="list-style-type: none"> <li>loaded</li> </ul>                | VOICE-003 | <input type="radio"/> not run here |
| <b>Notice bar</b><br>any main-process notice<br>windows-companion/src/renderer/views/NoticeBar.tsx    | <ul style="list-style-type: none"> <li>info</li> <li>warning</li> </ul> | VOICE-002 | <input type="radio"/> not run here |

## DISCORD BOT

Messages rendered from templates the operator can edit.

| SURFACE                                                                                                                                                     | STATES IT MUST ANSWER                                                                                                                                                                                                                                                                     | RULES                                              | SEEN                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|------------------------------------|
| <b>Bot replies</b><br>/play, /queue, /skip ... or the prefix form<br>docker-container/src/discord/gateway.ts<br>packages/domain/src/templates.ts<br>DEC-005 | <ul style="list-style-type: none"> <li>queued</li> <li>now playing</li> <li>skipped / vote counted</li> <li>refused (ephemeral)</li> <li>no results</li> <li>cannot be played here</li> <li>joined / left voice</li> <li>not in a voice channel</li> <li>missing Connect/Speak</li> </ul> | UX-DISCORD-001<br>UX-DISCORD-002<br>UX-DISCORD-003 | <input type="radio"/> not run here |

## ANDROID SHELL

A WebView around the player, plus system UI.

| SURFACE                                                                                                                                                                                                                                                 | STATES IT MUST ANSWER                                                                                                                                                                   | RULES       | SEEN                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------|
| <b>WebView shell</b><br>launcher icon<br>android/app/src/main/java/com/nowplaying/player/MainActivity.kt<br>android/app/src/main/res/values/strings.xml<br>DEC-012                                                                                      | <ul style="list-style-type: none"> <li>player loaded from assets</li> <li>renderer crashed (reloads)</li> <li>back = history, else background</li> <li>hub over http refused</li> </ul> | UX-SAFE-004 | <input type="radio"/> not run here |
| <b>Notification, picker and save dialog</b><br>a fetch starts; a file input opens; an export is saved<br>android/app/src/main/java/com/nowplaying/player/FetchService.kt<br>android/app/src/main/java/com/nowplaying/player/BlobDownloads.kt<br>DEC-010 | <ul style="list-style-type: none"> <li>fetching (foreground notification)</li> <li>permission prompt (API 33+)</li> <li>saved</li> <li>could not save</li> </ul>                        | VOICE-002   | <input type="radio"/> not run here |

## LOCAL HELPER

A console program that serves the player and runs download tools.



| SURFACE                                                                                                     | STATES IT MUST ANSWER                                                                                                            | RULES     | SEEN           |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------|----------------|
| <b>Local helper console</b><br>now-playing-helper<br>local-helper/src/cli.ts<br>local-helper/src/options.ts | <ul style="list-style-type: none"> <li>tools found / missing with install hint</li> <li>token shown</li> <li>stopping</li> </ul> | VOICE-002 | ○ not run here |

## Journeys

The ten things people come to do, from where they start to where they can fail. Each step list is what the code does today; each failure is one the product answers in words.

### First run of the hub

A fresh data volume; the container is up on 127.0.0.1:4546.

1. Open the admin page
2. Sign in with admin / admin
3. Choose a real password
4. Every section unlocks

#### WHEN IT GOES WRONG

- Weak password refused with the reason
- Login rate-limited after repeated failures

UX-SAFE-002

docker-container/tests/integration/setup-and-pairing.test.ts  
docker-container/tests/e2e/first-run.setup.ts

### Pair a player or the companion

Setup complete; the device can reach the hub.

1. Admin creates a pairing code (Devices)
2. Device enters the code
3. Both screens show the same fingerprint
4. Admin confirms
5. Device receives a scoped credential

#### WHEN IT GOES WRONG

- Code expired
- Fingerprint differs: do not confirm
- Wrong codes lock out that address for a while

UX-SAFE-003

docker-container/tests/integration/setup-and-pairing.test.ts

### Add music on this device and play it

The player is open (installed or not).

1. Library shows how to add music
2. Choose a folder or files
3. Tracks appear; unplayable formats say so
4. Play; the hero and the list mark the playing row

#### WHEN IT GOES WRONG

- Browser without folder access falls back to file choice
- A file that disappeared says it cannot be found

UX-STATE-001 UX-A11Y-001

music-player/tests/e2e/local-file.spec.ts  
music-player/tests/e2e/player.spec.ts

### Listen together

Player paired with a hub; a group exists.

1. Switch the mode to Shared
2. Join or create a group
3. Anyone queues; everyone hears it at the same point
4. Skips and pauses apply to the group

#### WHEN IT GOES WRONG

- No hub: the switch explains why
- Stale command: refused with the current revision
- Member without the DJ role: vote-skip instead of skip

UX-PRIN-002 UX-SHARED-001

docker-container/tests/integration/groups-and-queue.test.ts

## Play music into a Discord voice channel

Bot token installed, bot enabled, default group chosen, worker container running.

1. Join a voice channel
2. /play <search or link>
3. The bot joins your channel and replies Queued
4. Audio follows the group timeline; pause, skip and seek from any device apply

### WHEN IT GOES WRONG

- Not in a voice channel: the reply says so
- Missing Connect/Speak: the reply names them
- Source that cannot stream in Discord: skipped with the reason
- Worker not running: the admin panel says so

```
UX-DISCORD-001 UX-DISCORD-002 UX-DISCORD-003
docker-container/tests/integration/discord-
worker.test.ts
docker-container/tests/unit/discord-voice.test.ts
docker-container/tests/integration/discord-
parity.test.ts
```

## Download a track the hub may give you

Paired player; the provider allows it.

1. Row menu → Download
2. Choose format and the basis (e.g. content I own)
3. Progress in words
4. The file saves as a link download

### WHEN IT GOES WRONG

- Format unavailable without FFmpeg
- Provider refuses the basis
- Cancelled: the partial file is removed

```
UX-FEED-001 UX-LINK-001
docker-container/tests/security/ssrf-and-
downloads.test.ts
docker-container/tests/integration/downloads-
queue.test.ts
```

## Share a track or playlist

Paired player.

1. Share
2. Choose whether streaming is allowed
3. Copy the link
4. The recipient opens the public page

### WHEN IT GOES WRONG

- No hub: the sheet explains why
- Link revoked or used up: the page says it is unavailable

```
UX-PRIN-002
docker-container/tests/integration/shares-and-
sync.test.ts
docker-container/tests/security/share-limits.test.ts
```

## Send files from Windows to the hub

Companion paired; a folder indexed.

1. Add a folder
2. Select tracks in Music
3. Send to the hub
4. Transfers shows progress; cancel stops it

### WHEN IT GOES WRONG

- Drive not connected
- Hub unreachable: upload fails with the reason

```
UX-FEED-001
windows-companion/tests/integration/companion-and-
hub.test.ts
```

## Back up and restore

Setup complete.

1. Backup → Create
2. Restore → confirm
3. The hub reports success or the reason it failed

### WHEN IT GOES WRONG

- Invalid backup refused
- Swap failed: previous database put back

UX-SAFE-001

`docker-container/tests/integration/backup-and-search-cursor.test.ts`

## Fetch with yt-dlp or spotDL

The local helper is running, or the Android app has unpacked its tools.

1. Library → Fetch
2. Paste a link and pick a format
3. Progress
4. Files are added to the library

### WHEN IT GOES WRONG

- Tool missing: the control is not shown
- Tool error: its own message, without paths

UX-PRIN-001 VOICE-002

`music-player/tests/e2e/helper.spec.ts`

`local-helper/tests/integration/helper-server.test.ts`

## The Discord bot

The bot is the one surface Aqua does not draw: Discord does. What the suite controls is the words, the embed colour, whether a reply is private, and when the bot moves in and out of voice. Replies come from templates an operator can edit in the hub (Admin → Discord); these are the defaults, rendered with sample values by the same function the bot uses.

Commands: `/play` `/nowplaying` `/queue` `/skip` `/pause` `/resume` `/stop` `/shuffle` `/clear` `/join` `/leave` `/settings`. Every one also works as a prefix command (`!play`) once the Message Content intent is granted, and the two forms are identical by construction ([UX-DISCORD-001](#)). Embed colours are the system's own: selection blue, success, warning and danger.

*Illustration — Discord renders its own interface. This shows content and colour, not Discord's layout.*

|                                                                                                                       |                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>success</b><br>A command with nothing more specific to say                                                         | Done.                                                                                                         |
| <b>queued</b><br><code>/play</code> found something playable                                                          | <b>Queued #3</b><br><b>Signal Fade</b> — Cassette Bloom<br>Requested by alex · 0:04 · Public domain           |
| <b>nowPlaying</b><br><code>/nowplaying</code>                                                                         | <b>Now Playing</b><br><b>Signal Fade</b> — Cassette Bloom<br>Live from Pier 9<br>Requested by alex in Kitchen |
| <b>skipped</b><br><code>/skip</code> by a DJ or the requester                                                         | Skipped <b>Signal Fade</b> (skipped).                                                                         |
| <b>permissionDenied</b><br>The caller lacks the DJ or admin role<br>only the caller sees this                         | You don't have permission to do that (the DJ role is required).                                               |
| <b>noResults</b><br><code>/play</code> found nothing<br>only the caller sees this                                     | No results for “No results”.                                                                                  |
| <b>unavailableSource</b><br>The match cannot stream in Discord<br>only the caller sees this                           | <b>Signal Fade</b> cannot be played here: YouTube only plays in its own player.                               |
| <b>emptyQueue</b><br><code>/queue</code> or <code>/nowplaying</code> with nothing queued<br>only the caller sees this | The queue is empty. Use <code>/play</code> to add something.                                                  |
| <b>joined</b><br><code>/join</code> , or <code>/play</code> when the bot was not in voice                             | Joined #music.                                                                                                |
| <b>left</b><br><code>/leave</code> , or the idle timeout                                                              | Left the voice channel.                                                                                       |
|                                                                                                                       |                                                                                                               |

|                                                                                                   |                                               |
|---------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <b>error</b><br>Anything unexpected; the reason is the error message<br>only the caller sees this | Something went wrong: Queue is at revision 12 |
| <b>wrongChannel</b><br>A command outside the designated channel<br>only the caller sees this      | Music commands work in #music.                |
| <b>paused</b><br>/pause                                                                           | Paused.                                       |
| <b>resumed</b><br>/resume                                                                         | Resumed.                                      |
| <b>stopped</b><br>/stop                                                                           | Stopped and cleared the queue.                |
| <b>shuffled</b><br>/shuffle                                                                       | Shuffled 12 tracks.                           |
| <b>cleared</b><br>/clear                                                                          | Cleared 12 tracks from the queue.             |

### VOICE-CHANNEL REPLIES

Added to the reply when the bot cannot move into voice. They are sentences in the gateway, not templates.

- “Join a voice channel first (or use join <channel>), so I know where to play.”
- “I need View Channel, Connect and Speak in <channel> to play there.”
- “There is no voice channel called “<name>” that I can see.”
- “Could not connect to the voice channel. Check that the bot has Connect and Speak there, and that outbound UDP is not blocked.”

**Known gap (DEC-005).** The default `noResults` template quotes `{{reason}}`, but the command service passes the search text as `title`, so a failed search currently reads “No results for “No results”.”

# Platforms

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## WEB AND PWA — PLAYER AND HUB ADMIN

- **Breakpoints follow content, not a device list.** The page skin changes at 480, 560, 620, 720 and 900px; the window skin at 479, 599 and 759px, where the source list gives way and the current source is named in the toolbar. Check at the mockup sizes: 320, 390, 768, 1280 and 1680px.
- **The pointer decides density, not the width.** (`pointer: coarse`) switches the touch layer on at any width; a narrow desktop window keeps desktop density ([UX-TOUCH-001](#)).
- **Installable and offline.** The player ships a manifest, icons and a service worker, opens with the network off once it has loaded online, and loads nothing from another origin. The single-file build, `now-playing.html`, runs straight from disk.
- **Themes.** The page skin follows the system colour scheme; a page can pin itself light with `data-np-theme="light"`. The window skin is light only (DEC-003).
- **Keyboard.** Space toggles playback, the section strip and source lists move with the arrow keys, and Escape closes a popover without losing what was typed.

## ANDROID

- The app is the player inside a WebView, loaded from its own assets, so every web rule applies unchanged. The player is built with `NP_BASE_PATH=/assets/app/`.
- Back goes back in the page's history; with nowhere to go it moves the app to the background, so music keeps playing.
- System UI carries its own copy (`strings.xml`): the "Fetching music" notification while a download runs, the document picker for file inputs, and the save dialog for exports ("Saved ..." / "Could not save ..."). Notification permission is asked for the first time a fetch starts (DEC-010).
- Hubs are reached over https only (DEC-012).

## WINDOWS COMPANION — ELECTRON

- One window with a source list, and a tray icon (`resources/tray.svg`) whose menu has Open and Quit. With the "minimize to tray" preference on, closing the window keeps the companion running.
- It is the only product that reads the filesystem; its copy says so and never implies files are copied or uploaded in the background.
- Pairing shows the hub's fingerprint and warns when the hub address is plain http off the local network.

## TERMINAL — LOCAL HELPER

- Takes the first free port in a small range, prints the token to paste into Settings → Platforms, and names each missing tool with the command that installs it.
- Ctrl+C removes only the helper's own working folder, never the folder it was pointed at.

## ENGINEERING APPENDIX — DOCKER DELIVERY

Docker is how the hub is delivered, not a design surface. What it means for a design change:

- The image (`docker-container/Dockerfile`) builds the admin GUI with the same `@now-playing/aqua-ui`, so a token or component change reaches the hub only when the image is rebuilt: `docker compose up -d --build` OR `./nowplaying update`.
- The hub listens on port 4546, published to `127.0.0.1` by default, with `/healthz` for the health check. Host names other than IP addresses, `localhost` and `.local/.lan` names must be listed in `NP_ALLOWED_HOSTS` or set as `NP_PUBLIC_ENDPOINT`.
- The Discord worker is the same image under `--profile discord`. Its default replies change when `packages/domain/src/templates.ts` changes and the image is rebuilt, or immediately when an operator edits a template in the hub.
- The player, the Android app and the companion are built separately; each needs its own rebuild to pick up a design change.

## Every product, every size — and editable

---

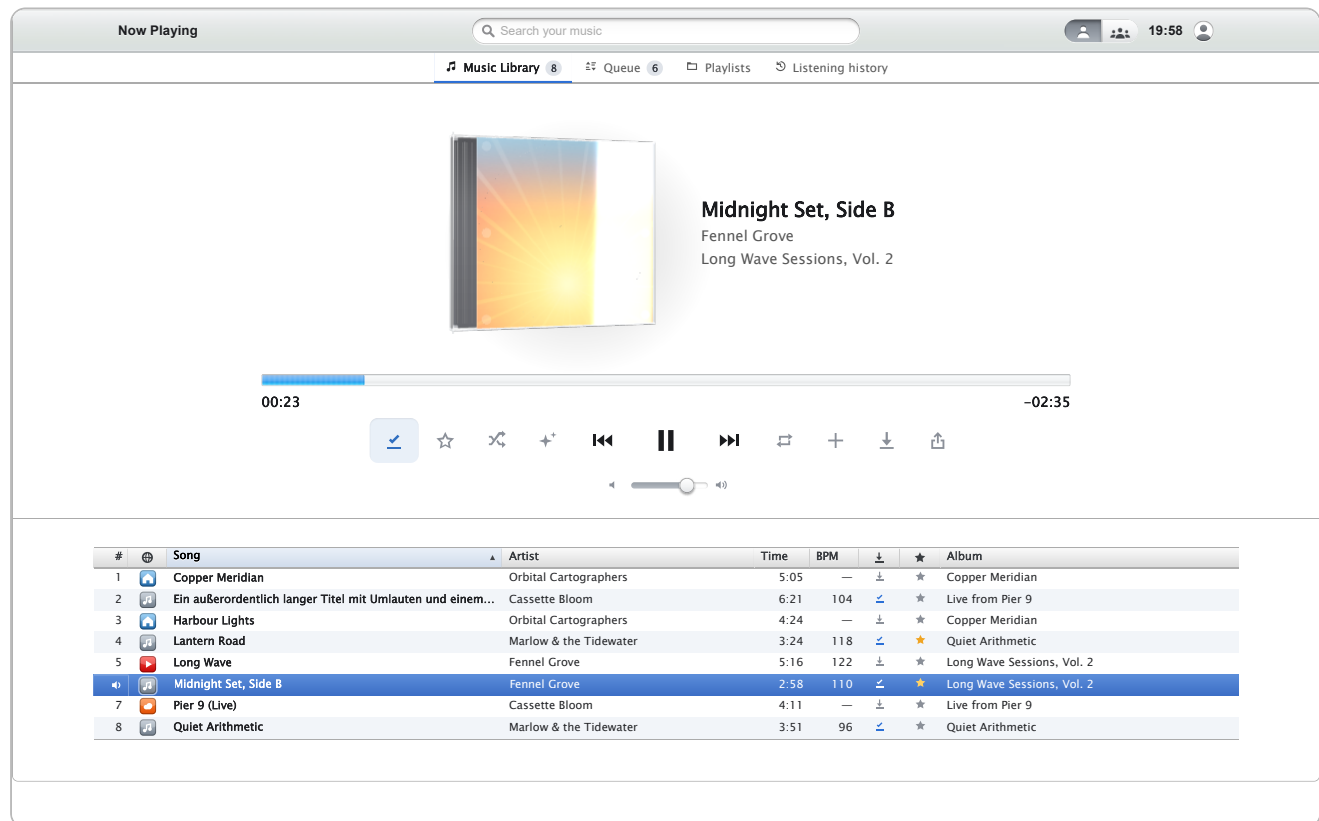
Each frame below is an iframe with its own viewport, so a 320px column is a real 320px viewport and this system's media queries behave exactly as they do on a phone. The screens inside are built from the components on this page, so they are 1:1 with the products by construction rather than by somebody keeping two drawings in step. Under each one is a measurement taken *inside* that frame: what runs off the side, the smallest text, the smallest thing you could tap.

A frame marked *touch layer emulated* has this system's (`pointer: coarse`) rules re-applied inside it, read back out of the real stylesheets — an iframe inherits the desktop's pointer, so without that a phone frame would quietly show desktop sizes. At the bottom you can edit any custom property the stylesheets define and watch every frame repaint at once; what you copy out is a block the three products already read.

In print, every screen is shown at the laptop and phone sizes with the fit measured inside each frame. The interactive version — every size, the product switch and the token editor — is <docs/design/styleguide.html>.

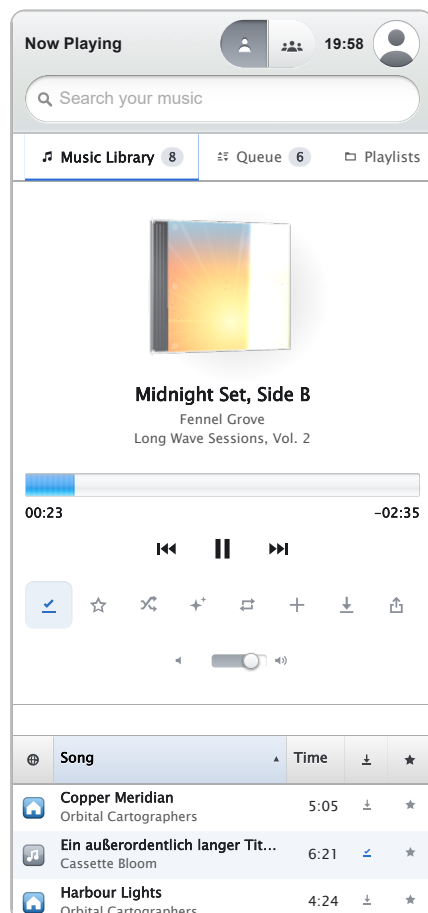
## Now playing and library

The whole page skin at once. Narrow it and watch the list drop its number and artist columns and put the artist under the title instead.



Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 10px



Phone 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 12px · smallest target 44px



## Settings

Where most of the app's explaining happens, so it is the screen where text size and wrapping matter most.

**Settings** What this app can do here, and what it cannot — with the reason in each case.

**Playback**

☒ Crossfade songs

☒ Keep songs from the same album gapless

**Platforms**

Three questions, because the platforms answer them differently. Every "no" says whose rule it is.

**This device** ☒ Play here: Yes ☒ Keep offline: Yes ☒ Save a file: Yes

**Bandcamp** ☐ Play here: No ☐ Keep offline: Sometimes ☐ Save a file: Sometimes

**YouTube** ☐ Play here: Sometimes ☐ Keep offline: No ☐ Save a file: No

**Spotify** ☐ Play here: Sometimes ☐ Keep offline: No ☐ Save a file: No

**Hub**

Hub address

https://living-room.local:4546

**Laptop** 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 12px

**Settings** What this app can do here, and what it cannot — with the reason in each case.

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**Spotify** ☐ Play here: Sometimes ☐ Keep offline: No ☐ Save a file: No

**Hub**

Hub address

https://living-room.local:4546

**Phone** 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 13px · smallest target 44px

Now playing (audio chain)

What the audio chain is doing to the song, and the spectrum visualiser under it — drawn by the player’s own spectrum code from a generated analyser frame.

Now playing

What the audio chain is doing to this song, measured rather than assumed.

How this is being played

Song

Midnight Set, Side B — Fennel Grove

Position

0:23 of 2:58

Source

A file on this device

Equalizer

Bass Lift — the preset chosen for this album

Limiter

On — prevents clipping when the equalizer adds gain

Headroom

3.0 dB trim applied to keep the boosted signal below full scale

Retuning

Off — playing at the original pitch

Added latency

11.6 ms



Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 11px

Now Playing styleguide · source 4e43856fb479381b

50 / 63

Now playing

What the audio chain is doing to this song, measured rather than assumed.

How this is being played

Song

Midnight Set, Side B — Fennel Grove

Position

0:23 of 2:58

Source

A file on this device

Equalizer

Bass Lift — the preset chosen for this album

Limiter

On — prevents clipping when the equalizer adds gain

Headroom

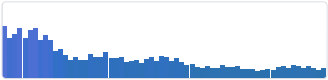
3.0 dB trim applied to keep the boosted signal below full scale

Retuning

Off — playing at the original pitch

Added latency

11.6 ms

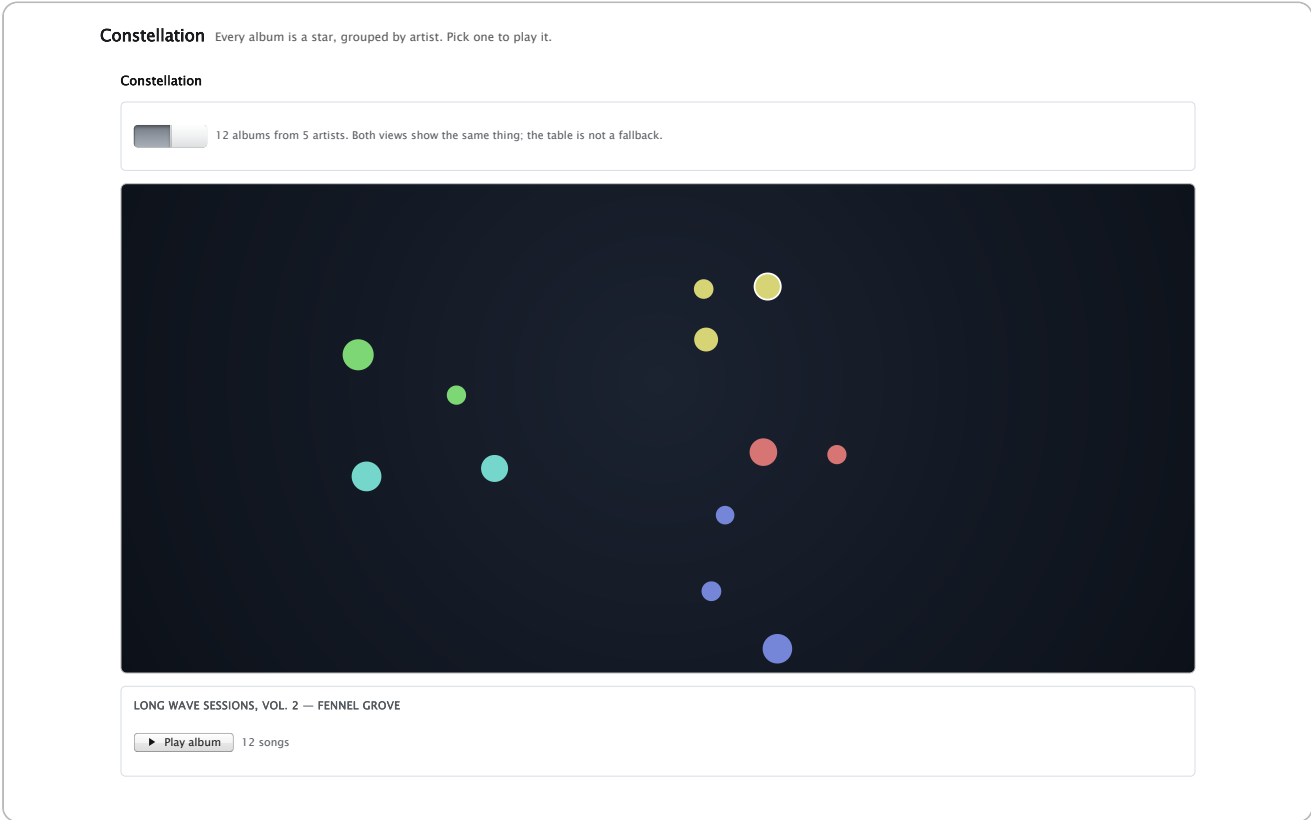


Phone 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 13px · smallest target —

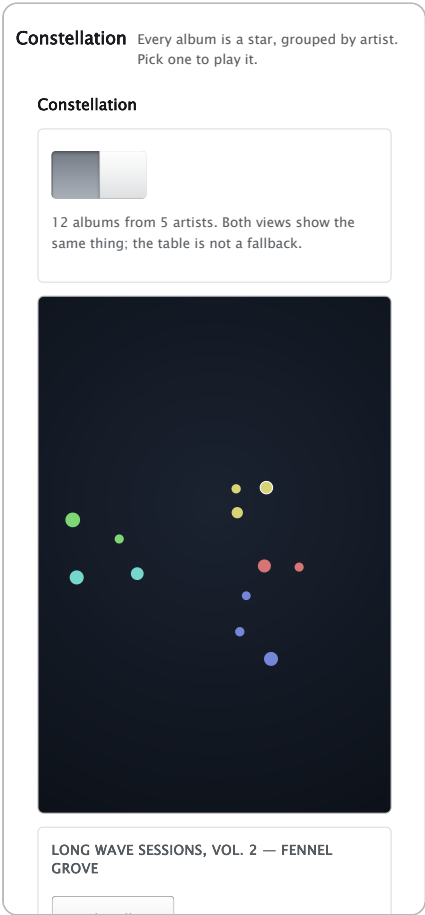
Constellation

The library as a star field, placed by the player’s own layout code. The table beside it is the same information, not a fallback.



Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 11px

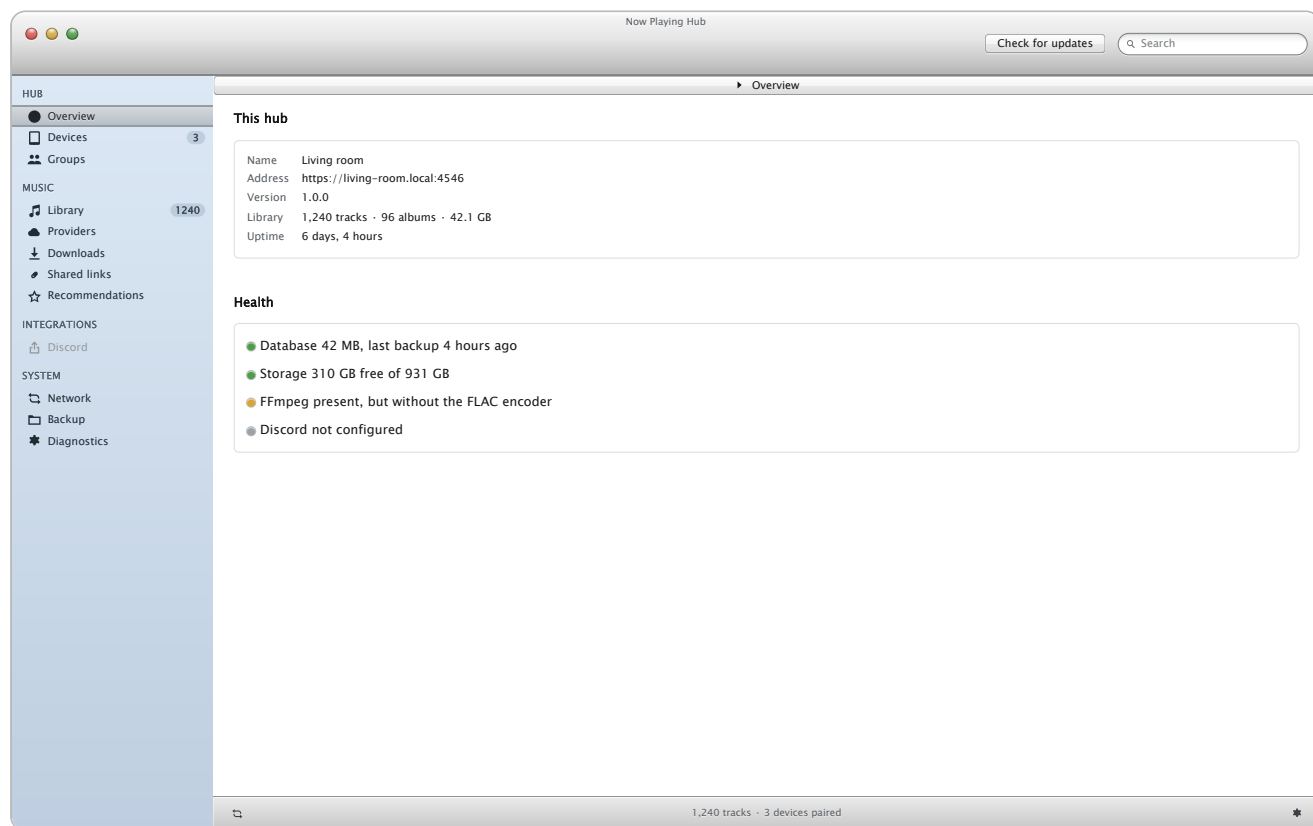


Phone 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 13px · smallest target 44px

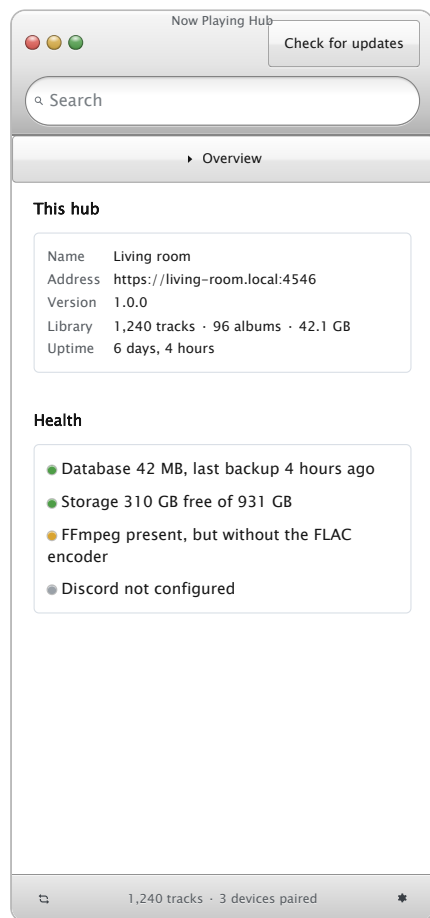
## Overview

The window skin with a source list. Below about 700px the sidebar gives way and the current source is named in the toolbar instead.



**Laptop** 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 10px

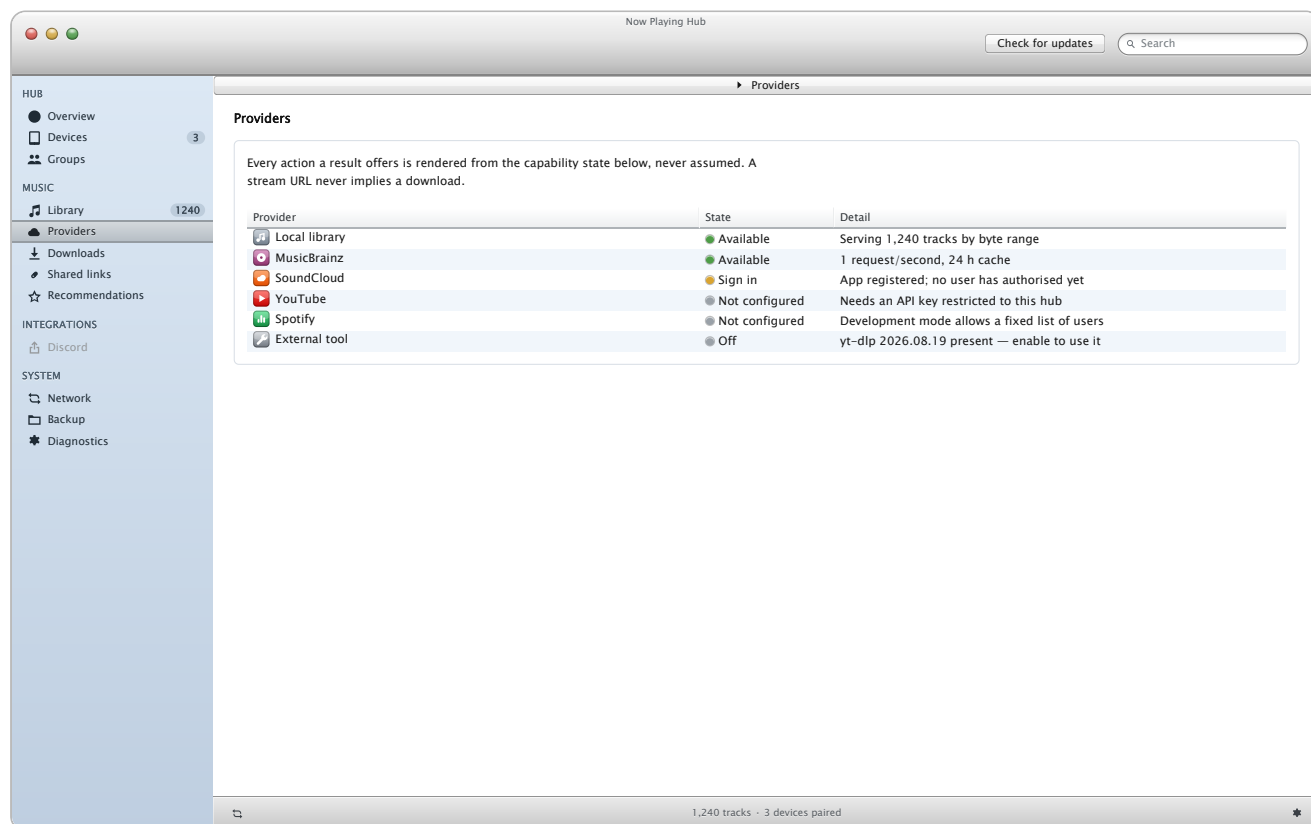


**Phone** 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 12px · smallest target 44px

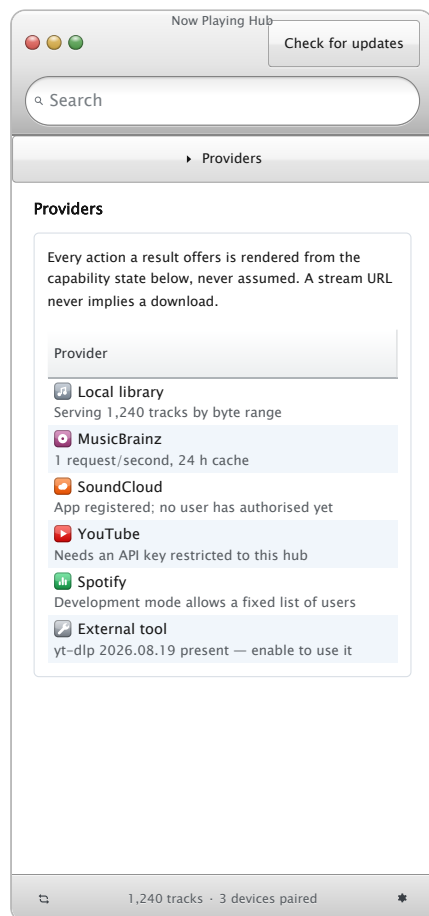
## Providers

A table inside a window. On a phone the table stacks rather than scrolling sideways.



**Laptop** 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 10px

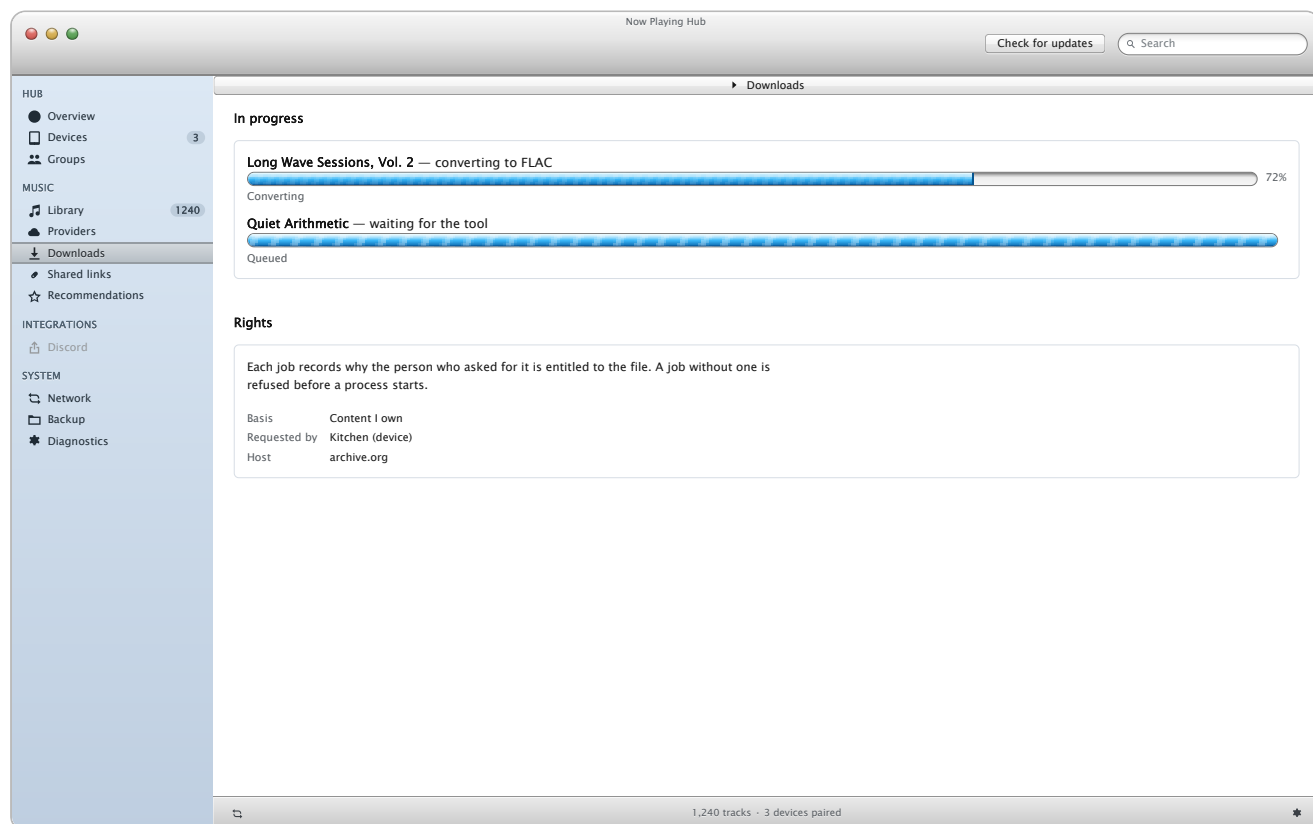


**Phone** 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 12px · smallest target 44px

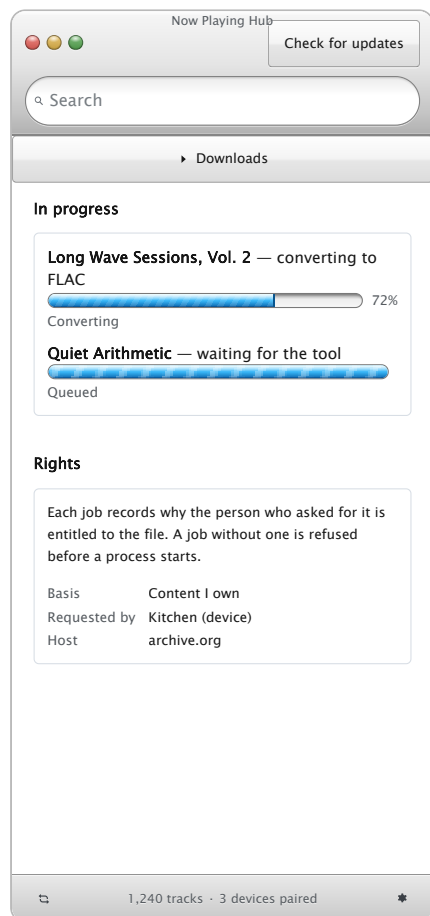
## Downloads

Progress and a rights basis. Both are text that has to stay readable when the window is half its width.



**Laptop** 1280 × 800, shown at 53%

● Fits · nothing off the side · smallest text 10px

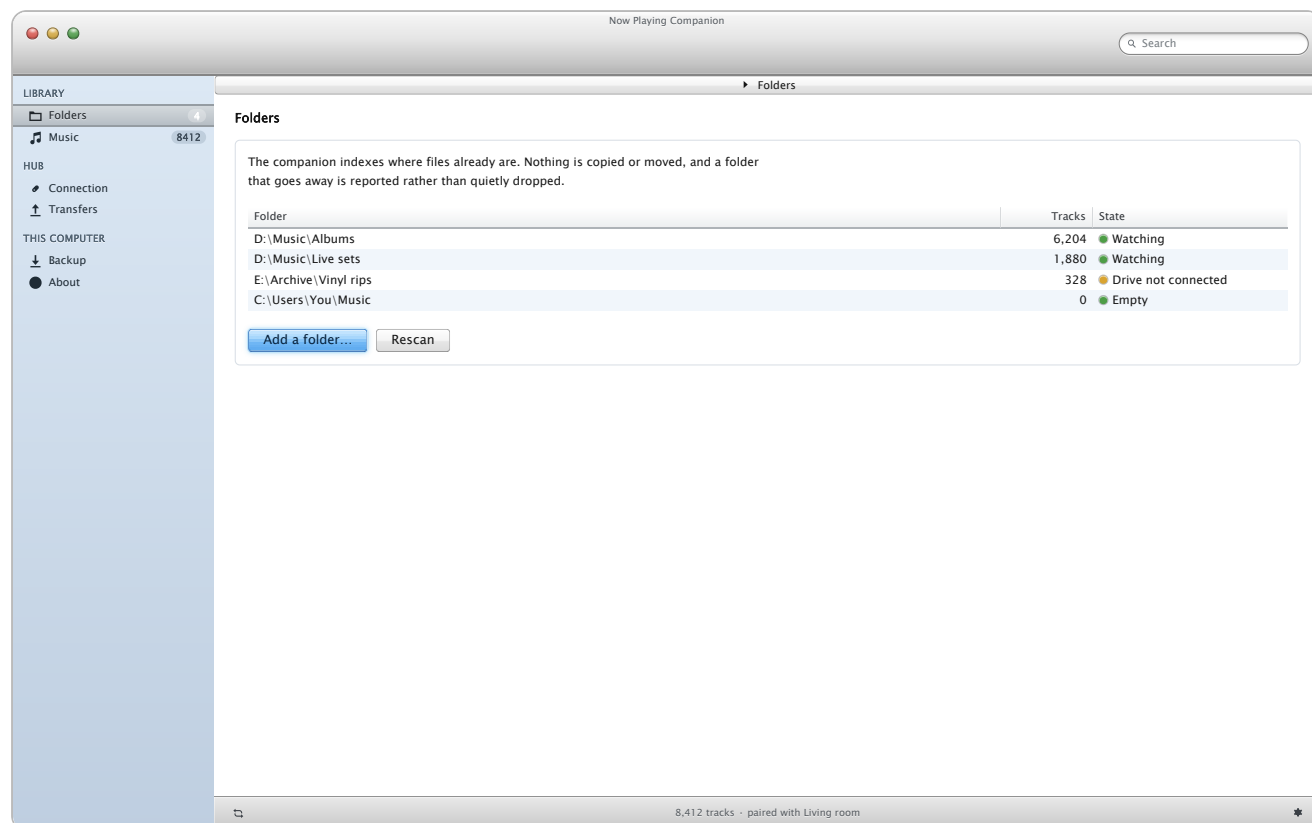


**Phone** 390 × 844, shown at 56% touch layer emulated

● Fits · nothing off the side · smallest text 12px · smallest target 44px

## Folders

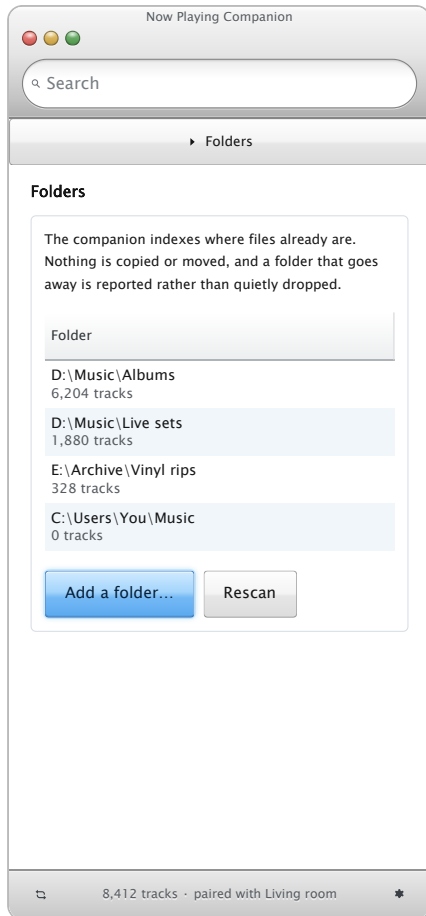
The same window skin the hub uses, on a desktop where it is never narrow — which is exactly why it is worth checking that it still holds up when it is.



Laptop 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 10px



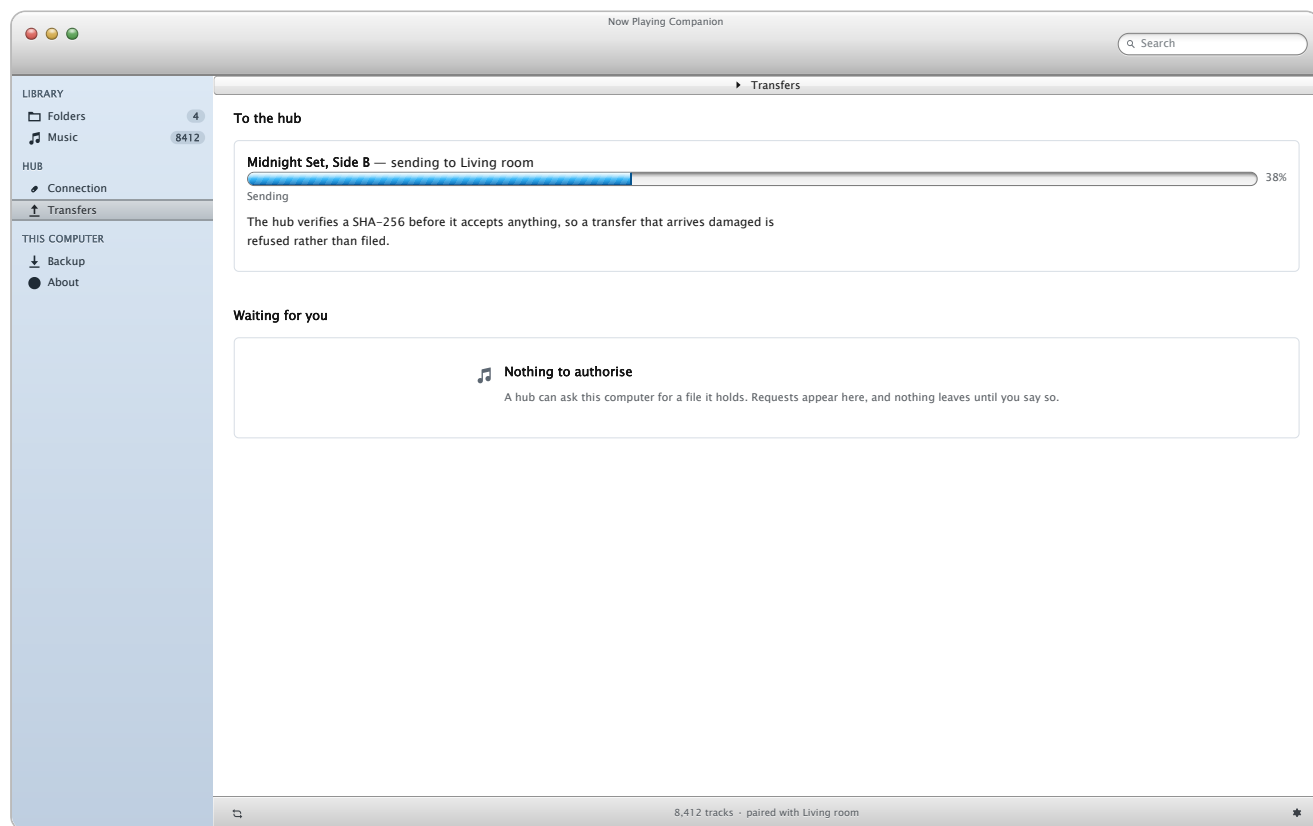


**Phone** 390 × 844, shown at 56% touch layer emulated

● Fits · nothing off the side · smallest text 12px · smallest target 44px

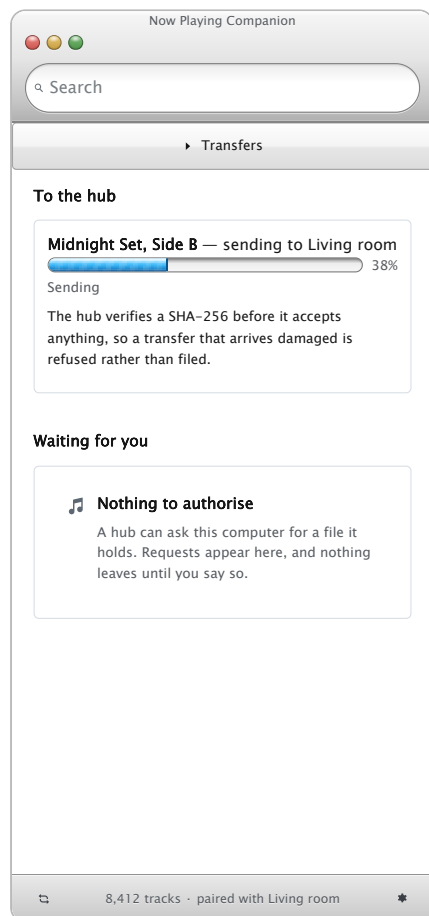
## Transfers

Progress, and an empty state that has to say something useful rather than apologise.



**Laptop** 1280 × 800, shown at 53%

● Fits nothing off the side · smallest text 10px



**Phone** 390 × 844, shown at 56% touch layer emulated

● Fits nothing off the side · smallest text 12px · smallest target 44px

## Sources and commands

This page is generated. The sources below are the design authority: change them, then rebuild. The exports — this page and its PDF — are views, so an edit or annotation on either is a request to change a source, not a change in itself.

| CONCERN       | SOURCE                                                                                                                                                                                                                               | OWNS                                                                                                                                                                                   |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tokens        | <code>packages/aqua-ui/src/styles/tokens.json</code>                                                                                                                                                                                 | Documented values: colour, type, space, size, shadow, motion, page skin.                                                                                                               |
| stylesheets   | <code>packages/aqua-ui/src/styles/aqua.css</code><br><code>packages/aqua-ui/src/styles/aqua-window.css</code><br><code>packages/aqua-ui/src/styles/aqua-media.css</code><br><code>packages/aqua-ui/src/styles/now-playing.css</code> | The custom properties every product reads (light scheme on <code>:root</code> , dark and touch layers after it). Mirrors <code>tokens.json</code> by hand; see <code>tokenMap</code> . |
| tokenMap      | <code>design/token-map.json</code>                                                                                                                                                                                                   | Which custom property carries each token, and recorded exceptions.                                                                                                                     |
| overrides     | <code>packages/aqua-ui/src/styles/overrides.css</code>                                                                                                                                                                               | Temporary cross-product trials written by the styleguide's token editor. Empty in the repository.                                                                                      |
| components    | <code>packages/aqua-ui/src</code>                                                                                                                                                                                                    | Every shared control, shell, overlay and state. Imported by the player, the hub admin GUI and the Windows companion.                                                                   |
| rules         | <code>design/ux-rules.json</code>                                                                                                                                                                                                    | Interaction and content rules with stable IDs, their owning code and their tests.                                                                                                      |
| coverage      | <code>design/coverage.json</code>                                                                                                                                                                                                    | Every surface, state and journey, with discovery sources.                                                                                                                              |
| decisions     | <code>design/decisions.md</code>                                                                                                                                                                                                     | Adopted, proposed and unresolved decisions and exceptions (DEC-###).                                                                                                                   |
| prose         | <code>docs/design/STYLEGUIDE.md</code>                                                                                                                                                                                               | The reasoning behind the visuals, in long form.                                                                                                                                        |
| conformance   | <code>docs/AQUA_CONFORMANCE.md</code>                                                                                                                                                                                                | Spec §17/§18 MUSTs mapped to tests, reviewer checks or deviations.                                                                                                                     |
| specification | <code>docs/design/APPLE_AQUA_2009_2010_UI_DESIGN_SPEC.md</code>                                                                                                                                                                      | The reconstruction target (Snow Leopard / iTunes 9).                                                                                                                                   |

### CHANGING SOMETHING

1. Find the owner: a token, a component, a rule by ID, or a surface in the ledger.
2. Change it there — and, for a rule, the code that implements it and the test that proves it.
3. Run `pnpm styleguide:check` and the tests the rule names; then `pnpm styleguide:build` and `pnpm styleguide:pdf`.
4. Look at the result in the mockups at every size, and at every page of the PDF.
5. A deliberate exception goes in `design/decisions.md` with its reason. Never widen an exception to make a check pass.

### COMMANDS

**STYLEGUIDE:BUILD** rebuilds `docs/design/styleguide.html` from the sources above (same as `pnpm build:styleguide`).

**D**

**STYLEGUIDE:CHECK** sources and evidence exist, rule IDs are unique, every discovered screen is in the ledger, `tokens.json` matches the stylesheets, and the committed HTML was built from the current sources.

|                |                                                                                                                                         |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| STYLEGUIDE:PDF | prints the committed HTML to docs/design/styleguide.pdf with Playwright's Chromium (set NP_STYLEGUIDE_BROWSER to use another Chromium). |
| E2E            | measures every mockup at every size.                                                                                                    |
| CONFORMANCE    | includes packages/aqua-ui/tests/unit/aqua-conformance.test.ts and design-sources.test.ts.                                               |

## Changing any of this

---

- **Tokens first.** A new colour or size goes in `tokens.json`, then into the stylesheet as a custom property. A literal hex in a component is a bug.
  - **Both schemes.** Light on bare `:root`; redefine only what changes inside the dark guard.
  - **Both skins, if it is a control.** Button, Slider and friends are shared by three products.
  - **Say why in the CSS.** Every unobvious value carries a comment explaining what it reconstructs.
  - **The gates.** `AQUA_CONFORMANCE.md` maps every §17/§18 MUST to a test, a named reviewer check, or a recorded deviation, and `pnpm verify` runs the lot. A deviation is legitimate; an undocumented one is not.
  - **Rebuild this page.** `pnpm styleguide:build` regenerates `docs/design/styleguide.html` from the same source the products use, `pnpm styleguide:check` fails if it no longer matches its sources, and `pnpm styleguide:pdf` prints it.
- 

The correction log in `DEVIATIONS.md` is worth reading before a large change. It records the mistakes this system has already made — buttons a decade too glossy, a list redesigned instead of expanded, a picker that indexed music it could never play — and every one was found by building the thing and looking at it.

## Coverage and decisions

What this guide covers and how each part was checked, for source revision 4e43856fb479381b. The denominator is every surface in `design/coverage.json`, which the check keeps in step with the products' navigation.

| PRODUCT                    | SURFACES | IN<br>MOCKUPS | SEEN<br>RUNNING | NOT RUN BECAUSE                                                                             |
|----------------------------|----------|---------------|-----------------|---------------------------------------------------------------------------------------------|
| Music player               | 17       | 5             | 7               | —                                                                                           |
| Hub admin and public pages | 17       | 3             | 1               | —                                                                                           |
| Windows companion          | 7        | 2             | 0               | Electron-only; its renderer needs the preload bridge. Covered by DOM tests and the mockups. |
| Discord bot                | 1        | 0             | 0               | Needs a bot token and a Discord server; covered by the voice and parity tests.              |
| Android shell              | 2        | 0             | 0               | No Android SDK or JDK 17 on the machine that built this revision.                           |
| Local helper               | 1        | 0             | 0               | Console output was not captured for this revision; covered by its integration tests.        |

Automatic checks when this page was built: 25 rules with evidence, 34 surfaces discovered from navigation, 12 Discord commands, 177 tokens compared with the stylesheets (1 recorded exceptions). No open problems.

recorded exception: `font.family` ↔ `--aqua-font: Lucida Grande, Lucida Sans Unicode, Helvetica Neue, Arial, sans-serif` vs "Lucida Grande", "Aqua Fallback", "Lucida Sans Unicode", "Helvetica Neue", Arial, sans-serif (DEC-002)

### SEEN IN A RUNNING PRODUCT

| WHEN       | SCENARIO                                                                                                                                                                                                                                                                                           | VIEWPORTS ·<br>THEME                  | SURFACES |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|----------|
| 2026-09-16 | Player dev server, empty library, no hub: each section, Settings, search with no match, the Shared segment (toast by pointer and keyboard), and one token change ( <code>--np-accent</code> ) seen in the player and the rebuilt guide, then reverted.<br>Claude browser pane with DOM measurement | 1024×768,<br>375×812 touch ·<br>light | 7        |
| 2026-09-16 | Hub admin dev server against a fresh data folder: the first-run sign-in screen. Signing in was not attempted; it needs a person to enter the password.<br>Claude browser pane                                                                                                                      | 1024×768 · light                      | 1        |

### DECISIONS AND EXCEPTIONS

| ID      | STATUS    | DECISION                                                                                                                                                                      |
|---------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEC-001 | ● ADOPTED | Lucida Grande is not shipped (not redistributable). The window skin names it first and falls back through a tuned chain.                                                      |
| DEC-002 | ● ADOPTED | <code>--aqua-font</code> inserts the bundled "Aqua Fallback" face after Lucida Grande; <code>tokens.json font.family</code> does not list it. The CSS is authoritative.       |
| DEC-003 | ● ADOPTED | The window skin (hub admin, Windows companion) has no dark palette, because Snow Leopard had none. The page skin (player) and the styleguide follow the system colour scheme. |

| ID      | STATUS       | DECISION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEC-005 | ● UNRESOLVED | Discord <code>noResults</code> renders <code>No results for “{{reason}}”</code> , but the command service passes the search text as <code>title</code> and <code>No results</code> as <code>reason</code> , so the bot says “No results for “No results”.” Proposed fix: the template uses <code>{{title}}</code> . Changing a default template changes what existing servers see.                                                                                                                                                                                                        |
| DEC-006 | ● ADOPTED    | Cover Flow is not implemented; the 3D jewel-case stage stands in for it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| DEC-007 | ● ADOPTED    | The player replaces the inset LCD display with the hero, which states the same information at reading size.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| DEC-008 | ● UNRESOLVED | The public share page ( <code>docker-container/src/api/routes/shares.ts</code> ) is standalone HTML under a strict CSP with its own inline stylesheet. It uses a 12px pill button and weight 600, which the system otherwise forbids. Proposed: restyle it with the page-skin values (4px push button, weights 400/700).                                                                                                                                                                                                                                                                  |
| DEC-009 | ● ADOPTED    | The mockups render each product's real components with fixture data; they are not the product view files executed verbatim. Runtime coverage says which surfaces were seen in the running product and which only in a mockup.                                                                                                                                                                                                                                                                                                                                                             |
| DEC-010 | ● ADOPTED    | Android notifications, the document picker and the save dialog are system UI. The guide documents their copy ( <code>strings.xml</code> ) and when they appear, not their appearance.                                                                                                                                                                                                                                                                                                                                                                                                     |
| DEC-011 | ● PROPOSED   | <code>tokens.json</code> is mirrored into the stylesheets by hand; <code>pnpm styleguide:check</code> compares the two through <code>design/token-map.json</code> . Proposed: generate the custom properties from <code>tokens.json</code> so the map becomes unnecessary.                                                                                                                                                                                                                                                                                                                |
| DEC-012 | ● UNRESOLVED | The Android shell only reaches hubs over https (mixed content is never allowed and the manifest forbids cleartext), but the player's pairing form still accepts an <code>http://</code> address, so the refusal happens at connection time. Proposed: say so in the pairing form on Android.                                                                                                                                                                                                                                                                                              |
| DEC-013 | ● ADOPTED    | The Windows companion is Electron-only; its renderer cannot run in a browser without the preload bridge, so its runtime evidence comes from its DOM tests and the mockups.                                                                                                                                                                                                                                                                                                                                                                                                                |
| DEC-014 | ● ADOPTED    | The equalizer window ( <code>.eqw</code> ) has an intrinsic width: <code>max-width: 600px</code> , which holds its rails about 36 px apart the way the iTunes window does. It had been stretching to whatever container held it — 958 px in the styleguide — and read as a strip rather than a window.                                                                                                                                                                                                                                                                                    |
| DEC-015 | ● ADOPTED    | The note in every app icon is placed by centring the path's measured box (x 4.4–19.8, y 3–21.7 of its 24-unit grid) on the icon's optical target: the tile centre for the player and maskable icons, the folder mouth (240, 308) for the companion, the 32-square for the tray. The player icon had shipped 27 px left of centre; <code>tests/unit/icon-glyphs.test.ts</code> now pins all four placements.                                                                                                                                                                               |
| DEC-016 | ● ADOPTED    | Play sits on the progress rail's centre line at every width, with previous and next on its line either side. <code>KeyTransport</code> measures its row: the volume line stays beside the keys only while it fits on both sides (otherwise it goes beneath, centred), and the keys wrap only when they do not fit on one line, with previous, play and next alone on the first line. The player's eleven keys never leave room for the volume beside them inside the 820 px hero, so the player shows it beneath at every width. Covered by <code>tests/dom/page-chrome.test.tsx</code> . |
|         |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| ID      | STATUS    | DECISION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DEC-017 | ● ADOPTED | <p>The hub admin GUI is the six-tab window of <code>design/frontends/airwave-hub.html</code> (Overview, Devices, Music, Groups, Sharing, System); the thirteen sections keep their data flows and ledger entries and stack inside those tabs, and <code>#section-id</code> in the address still opens the right one. Two departures from that mockup stand. The first-run gate is a screen, not a banner over a disabled interface: the server refuses every gated route until the password is changed, so a live-looking interface behind the banner would be a picture of one. And the invite table shows who an invite is for and its state, not its code, because the hub keeps only a code's hash; the code appears once, when made, and the link needs the address players open Now Playing at. The mockup's <code>warning</code> (<code>#d99a1e</code>) yields to the existing <code>--aqua-warning</code>; its header and gate colours joined <code>tokens.json</code> as <code>header*</code> and <code>warningBg/warningEdge</code>.</p> |

## NOT COVERED, ON PURPOSE

- `docker-deployment` — Docker is delivery, not a visual surface. Its ports, health check and how rebuilt assets reach users are in the engineering appendix of the styleguide.
- `styleguide` — The styleguide documents the products; it is not one of them.
- `aqua-ui-gallery` — Developer specimen page (`pnpm --filter @now-playing/aqua-ui dev`); its specimens are rendered in the styleguide.